Autodiductive

The sciences, constructed and observed

Camille Bartholdi

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Rules of notation

Mathematics grew in many places at once, and the same thing ended up with many notations. A map's value is written $f(x)$, fx, f_x or x^f, depending on the field and the century. Order is written with $\leq$ and with $\geq$. Products are written with a dot, a cross, or nothing at all. Each tradition reads easily to its own members and is a barrier to everyone else.

This book uses *one notation for one thing*. The rules below hold throughout, in both parts. Every notation is introduced once, in a numbered definition, and is then listed in the Register at the end of the book, grouped by the mathematical model it belongs to.

Rule 1 (One notation)**.** Every object, relation and operation has exactly one notation, and every notation has exactly one meaning. A notation is introduced in the numbered definition where its meaning is defined, and is recorded in the register.

Rule 2 (Abbreviations replace)**.** When a short notation is introduced for a longer expression, the short one is used from then on and the long one is not used again. Example: once $x \notin A$ is defined to mean $\neg(x \in A)$, the second form does not appear again.

Rule 3 (Maps)**.** For a map f, the statement "f sends x to y" is written

$$x \overset{f}{\longmapsto} y.$$

This is the only way values of maps appear. The value of a map is never written as a term: there is no $f(x)$, fx or f_x. A map is defined by saying what it sends each element to: "the map $f\colon A \to B$ with $x \overset{f}{\longmapsto} t$ for all $x \in A$". Sequences and families are maps, so the same rule applies to them (Definition 4.1.1).

Rule 4 (Operations)**.** An *operation* is a map that has been given a symbol of its own: $+$, $\cdot$, $\cup$, $\circ$, and so on. Its value *is* written as a term. A binary operation is written between its two arguments, as in $a + b$, and an operation of one argument by its symbol, as in $-a$ or a^{-1} (Definition 4.6.1). Products are always written with $\cdot$, never by putting letters side by side.

Rule 5 (Relations)**.** A relation is written between the two things it relates: $x \in A$, $x \leq y$, $x \sim y$, $x\ R\ y$ (Definition 3.3.1). Each relation has one symbol. In particular there is no $\geq$ and no $>$: $y \geq x$ is written $x \leq y$.

Rule 6 (Brackets)**.** Round brackets ( ) group, and form ordered pairs (a, b) (Definition 3.1.1). Curly brackets { } form sets. Square brackets [ ] form equivalence classes and images. Brackets never stand for applying a map.

Rule 7 (Predicates)**.** $P(x)$ stands for a statement in which the variable x occurs (Definition 1.4.1). It is a statement, not a map applied to x. Apart from predicates, round brackets after a symbol only group, as in $\mathcal{P}(A \cup B)$.

Rule 8 (Identification)**.** When a structure is copied into a larger one by an injective map that preserves all of its notation, the copy is identified with the original, and the same symbols are used for both. This happens for $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$ (Theorems 6.3.1 and 7.3.1). Likewise, when a new kind of object is built from an old one, it gets a name of its own: the number 0 is the empty set $\varnothing$, regarded as a number (Definition 5.1.3). The register marks every identification.

Rule 9 (Subscripts)**.** Subscripts only tell apart objects that have different names, such as x_1 and x_2, or attach a fixed set to a notation, as in id_A. A subscript never stands for the value of a map.

Rule 10 (Prose)**.** In prose, statements are said in words (“if and only if”, “for all”). Symbols are kept for formulas.

Part I

Deduction

Introduction to deduction

A deductive science does not look at the world. It fixes a small number of starting rules, the *axioms*, and constructs everything else from them. Whatever it proves is true *relative to those axioms*, without any observation. This is the sense in which a deductive science constructs its world.

The one rule

Nothing may be used before it has been constructed.

Every chapter uses only definitions and results from earlier chapters, or from earlier in the same chapter. When a later topic needs something new, the new thing gets its own definition first. So this part of the book is one single chain that can be read, and checked, from the beginning.

Where the chain starts

Something has to come first, and it can't be constructed from anything earlier. We start with:

(1) **Classical logic**, the rules of correct reasoning (Chapter 1). It is the language everything else is written in.

(2) **Zermelo–Fraenkel set theory**: one kind of object, the *set*, and one relation between objects, *membership* $\in$ (Chapter 2).

Everything after that is constructed: pairs, relations and maps, then the number systems

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R},$$

then sequences, functions, calculus, linear algebra, and from there the rest of mathematics.

Discrete mathematics first

Discrete mathematics deals with objects that are separate and can be counted: truth values, finite collections, whole numbers, fractions. It comes first because the rest of mathematics is built on it. The real numbers, and with them functions, series and calculus, are constructed from the rational numbers, and those are constructed in the first chapters:

$$\begin{array}{ccccccccccc} \text{logic} & \to & \text{sets} & \to & \text{pairs} & \to & \text{relations} & & \to & & \text{maps} \\ & & & & & & \downarrow & & & & \downarrow \\ \mathbb{Q} & \leftarrow & \mathbb{Z} & \leftarrow & \mathbb{N} & & \text{equivalence classes} & & & & \text{recursion} \end{array}$$

- A **natural number** is a particular set: $0 = \varnothing$, $1 = \{0\}$, $2 = \{0, 1\}$, and so on. Each number is the set of all numbers before it (Chapter 5).
- An **integer** is what a difference $a - b$ of natural numbers *would* be. Subtraction doesn't exist yet, so we use the pair (a, b) and treat two pairs as equal when they would give the same difference (Chapter 6).
- A **rational number** is built the same way from a quotient a/b of integers (Chapter 7).

Planned chapters, each using only what is already in place:

- finite and infinite sets, cardinality, Cantor's theorem;
- combinatorics: sums, products, binomial coefficients;
- divisibility and primes: Euclid's algorithm, positional notation, unique factorization;
- graphs;
- the real numbers, sequences and series;
- functions and calculus;
- linear algebra.

Conventions

- Numbered statements are counted per section: Definition 3.2.4 is the fourth numbered item of Section 3.2. Axioms, definitions, theorems, lemmas, corollaries and remarks all share one counter, so references are easy to find.
- A *theorem* is an important result, a *lemma* a stepping stone, a *corollary* a direct consequence.
- A reference such as "Lemma 5.2.2(3)" means item (3) of Lemma 5.2.2.
- Every proof ends with ■.
- Notation follows the Rules of notation: one notation for one thing, and every notation is listed in the Register.
- Exercises are optional. Nothing later depends on them.
- Examples and remarks may mention things that come later, to motivate or illustrate. Definitions and proofs never use them.

Chapter 1

Logic

Logic is the starting point, and it can't be constructed from anything earlier. It is the set of rules we agree to reason by. Everything after this chapter is built using these rules, and only these rules.

We use *classical logic*: every statement is either true or false, never both and never neither.

1.1 Propositions

Definition 1.1.1 (Proposition)**.** A *proposition* is a statement that is either true or false. Its *truth value* is T (true) or F (false).

For example, "every set is a subset of itself" is a proposition. "x is a subset of y" is not, since its truth depends on x and y. That kind of statement is handled in Section 1.4. We write propositions as $P, Q, R, \ldots$

1.2 Connectives

New propositions are formed from old ones with *connectives*. A connective is defined by its truth table: the truth value of the new proposition for every possible truth value of its parts.

Definition 1.2.1 (Connectives)**.**

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

- $\neg P$ (*not* P, negation);
- $P \wedge Q$ (P *and* Q, conjunction);
- $P \vee Q$ (P *or* Q, disjunction). This is the inclusive "or": it is true when both are true;
- $P \Rightarrow Q$ (*if* P *then* Q, implication). P is the *hypothesis* and Q the *conclusion*;
- $P \Leftrightarrow Q$ (P *if and only if* Q, equivalence).

Remark 1.2.2 (Vacuous truth). $P \Rightarrow Q$ only claims: *whenever* P *is true,* Q *is true as well.* It makes no claim when P is false, so it counts as true then (rows 3 and 4). This is called *vacuous truth*: "every element of the empty collection is red" is true, because no element can contradict it.

Definition 1.2.3 (Tautology, contradiction, equivalence)**.** A compound proposition is a *tautology* if it is true for every assignment of truth values to its parts, and a *contradiction* if it is false for every assignment. Two compound propositions A and B are *logically equivalent*, written $A \equiv B$, if $A \Leftrightarrow B$ is a tautology, so they have the same truth table.

Theorem 1.2.4 (Basic laws)**.** *For all propositions P, Q, R:*

Law	*Statement*
Double negation	$\neg\neg P \equiv P$
Excluded middle	$P \vee \neg P$ *is a tautology*
Non-contradiction	$P \wedge \neg P$ *is a contradiction*
Commutativity	$P \wedge Q \equiv Q \wedge P, \quad P \vee Q \equiv Q \vee P$
Associativity	$(P \wedge Q) \wedge R \equiv P \wedge (Q \wedge R), \quad$ *same for* $\vee$
Distributivity	$P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$
	$P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$
De Morgan	$\neg(P \wedge Q) \equiv \neg P \vee \neg Q, \quad \neg(P \vee Q) \equiv \neg P \wedge \neg Q$
Implication as "or"	$P \Rightarrow Q \equiv \neg P \vee Q$
Contrapositive	$P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$
Negated implication	$\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$
Equivalence	$P \Leftrightarrow Q \equiv (P \Rightarrow Q) \wedge (Q \Rightarrow P)$

Proof. Each law is checked by writing out the truth table of both sides for all 2, 4 or 8 assignments and comparing them row by row. We show one case, De Morgan's first law:

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$
T	T	T	**F**	F	F	**F**
T	F	F	**T**	F	T	**T**
F	T	F	**T**	T	F	**T**
F	F	F	**T**	T	T	**T**

The bold columns agree. The other laws are checked the same way. ■

Remark 1.2.5. The *converse* $Q \Rightarrow P$ is *not* equivalent to $P \Rightarrow Q$. Take P false and Q true: then $P \Rightarrow Q$ is true and $Q \Rightarrow P$ is false.

1.3 Valid inference

Definition 1.3.1 (Valid inference)**.** An inference from premises $A_1, \ldots, A_n$ to a conclusion B is *valid* if $(A_1 \wedge \cdots \wedge A_n) \Rightarrow B$ is a tautology. In that case, whenever all premises are true, the conclusion is true too.

Theorem 1.3.2 (Basic inference rules)**.** *The following inferences are valid:*

Rule	*Premises*	*Conclusion*
Modus ponens	$P, \;\; P \Rightarrow Q$	Q
Modus tollens	$\neg Q, \;\; P \Rightarrow Q$	$\neg P$
Chaining	$P \Rightarrow Q, \;\; Q \Rightarrow R$	$P \Rightarrow R$
Cases	$P \vee Q, \;\; P \Rightarrow R, \;\; Q \Rightarrow R$	R
Contradiction	$\neg P \Rightarrow (Q \wedge \neg Q)$	P

Proof. Again by truth tables. For modus ponens: $(P \wedge (P \Rightarrow Q)) \Rightarrow Q$ could only be false if Q is false and $P \wedge (P \Rightarrow Q)$ is true. Then P is true, and $P \Rightarrow Q$ with P true and Q false is false. That is a contradiction, so the implication is never false. The others are similar. ■

1.4 Predicates and quantifiers

Definition 1.4.1 (Predicate)**.** A *predicate* is a statement containing a *variable* x, which becomes a proposition once a specific object is put in for x. We write it $P(x)$, and $P(a)$ for the proposition obtained by putting in a. Predicates can contain several variables: $P(x, y)$.

Definition 1.4.2 (Quantifiers)**.**

- $\forall x\ P(x)$ ("for all x, $P(x)$") is true if and only if $P(a)$ is true for every object a.
- $\exists x\ P(x)$ ("there exists x with $P(x)$") is true if and only if $P(a)$ is true for at least one object a.
- $\exists! x\ P(x)$ ("there is exactly one x with $P(x)$") abbreviates

$$\exists x\ (P(x) \wedge \forall y\ (P(y) \Rightarrow y = x)).$$

The variable after a quantifier is *bound*: $\forall x\, P(x)$ and $\forall y\, P(y)$ say the same thing. A variable that isn't bound is *free*.

Definition 1.4.3 (Equality)**.** Our logic includes the symbol $=$. The statement $a = b$ means a and b are the same object. It satisfies $a = a$, and if $a = b$ then anything true of a is true of b (*substitution*). We write $a \neq b$ for $\neg(a = b)$.

From the two rules of equality: $a = b \Rightarrow b = a$, and $(a = b \wedge b = c) \Rightarrow a = c$.

Definition 1.4.4 (Defining new notation)**.** $A :\Leftrightarrow B$ means that the new statement A is defined to mean the statement B. $a \coloneqq t$ means that the new name a is defined to be the object t.

Theorem 1.4.5 (Negating quantifiers)**.**

$$\neg\forall x\ P(x)\ \equiv\ \exists x\ \neg P(x), \qquad \neg\exists x\ P(x)\ \equiv\ \forall x\ \neg P(x).$$

Proof. $\forall x\, P(x)$ is false exactly when some object a makes $P(a)$ false, which means $\neg P(a)$ is true for some a. That is the first equivalence. For the second, apply the first to the predicate $\neg P$ and use double negation: $\neg\forall x\, \neg P(x) \equiv \exists x\, \neg\neg P(x) \equiv \exists x\, P(x)$. Negating both sides gives the claim. ■

For example, the negation of "every set has an element" is "some set has no element".

Remark 1.4.6 (Order of quantifiers)*.* Quantifiers of the same kind can be swapped,

$$\forall x\, \forall y\ P(x, y)\ \equiv\ \forall y\, \forall x\ P(x, y),$$

and the same holds for $\exists$. Mixed quantifiers cannot. In general

$$\exists y\, \forall x\ P(x, y)\ \Rightarrow\ \forall x\, \exists y\ P(x, y)$$

holds, but the converse doesn't. "For every number there is a bigger one" is true for the natural numbers. "There is a number bigger than every number" is false.

Definition 1.4.7 (Bounded quantifiers)**.**

$$\forall x \in A\ P(x) :\Leftrightarrow \forall x\, (x \in A \Rightarrow P(x)), \qquad \exists x \in A\ P(x) :\Leftrightarrow \exists x\, (x \in A \wedge P(x)),$$

and $\exists! x \in A\ P(x)$ means there is exactly one x with $x \in A$ and $P(x)$. Several variables are bound at once by $\forall x, y \in A$.

By Theorem 1.4.5, $\neg\forall x \in A\, P(x) \equiv \exists x \in A\, \neg P(x)$, and similarly for $\exists$. (The relation $\in$ is introduced in Chapter 2.)

1.5 Proof methods

A *proof* is a finite chain of statements. Each one is an axiom, a definition, a result already proved, an assumption made for the proof, or follows from earlier statements by a valid inference. These are the standard patterns:

Direct proof
of $P \Rightarrow Q$: assume P, derive Q.

Contrapositive
of $P \Rightarrow Q$: assume $\neg Q$, derive $\neg P$. Valid by Theorem 1.2.4.

Contradiction
for P: assume $\neg P$, derive some statement together with its negation. Valid by Theorem 1.3.2.

Cases:
if $P_1 \vee \cdots \vee P_n$ holds and R follows from each P_i, then R holds.

Equivalence
$P \Leftrightarrow Q$: prove $P \Rightarrow Q$ and $Q \Rightarrow P$ separately.

Universal statement
$\forall x\, P(x)$: let x be *arbitrary*, assuming nothing about it, and prove $P(x)$.

Existence
$\exists x\, P(x)$: name a specific object a and prove $P(a)$. Alternatively, prove it by contradiction.

Uniqueness
in $\exists! x\, P(x)$: prove existence, then assume $P(x)$ and $P(y)$ and derive $x = y$.

Induction is added in Chapter 5. There it is a theorem, proved from the axioms, not a new rule of logic.

Exercises

1. Show by truth table that $(P \Rightarrow Q) \wedge (P \Rightarrow R) \equiv P \Rightarrow (Q \wedge R)$.
2. Is $(P \Rightarrow Q) \Rightarrow R$ equivalent to $P \Rightarrow (Q \Rightarrow R)$? Prove it or give a counterexample.
3. Write the negation of $\forall \varepsilon\, \exists \delta\, \forall x\ (P(x, \delta) \Rightarrow Q(x, \varepsilon))$ without a $\neg$ in front of a quantifier. This shape comes back in calculus.

Chapter 2

Sets

We now fix the world that logic talks about. It contains a single kind of object, *sets*, and a single basic relation between them, *membership*.

2.1 Membership

Definition 2.1.1 (Membership)**.** For objects x and A, $x \in A$ is a proposition: "x is an element of A". We write $x \notin A$ for $\neg(x \in A)$.

There is nothing else: no numbers, no points, no symbols. Every mathematical object in this book, including numbers, pairs, maps and later functions and vectors, will be *a set built in a specific way*.

What we may assume about sets is fixed by the axioms of *Zermelo–Fraenkel set theory* (ZF). They are introduced one at a time, each where it is first needed.

2.2 Equality of sets

Axiom 2.2.1 (Extensionality)**.** Two sets with the same elements are equal:

$$\forall A\, \forall B\, \big(\forall x\, (x \in A \Leftrightarrow x \in B) \Rightarrow A = B\big).$$

So a set is *nothing more* than its elements. Order and repetition don't count: $\{a, b\} = \{b, a\} = \{a, a, b\}$, once these are defined (Axiom 2.5.1).

Definition 2.2.2 (Subset)**.** $A \subseteq B :\Leftrightarrow \forall x\, (x \in A \Rightarrow x \in B)$. We say A is a *subset* of B. A *proper subset*, $A \subsetneq B$, is one with $A \subseteq B$ and $A \neq B$.

Theorem 2.2.3. *For all sets A, B, C:*

(1) $A \subseteq A$;

(2) if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$;

(3) $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Proof. (1) For any x, $x \in A \Rightarrow x \in A$ is a tautology.

(2) Let $x \in A$. Then $x \in B$ because $A \subseteq B$, and then $x \in C$ because $B \subseteq C$ (chaining, Theorem 1.3.2).

(3) If $A = B$, then (1) gives both inclusions by substitution. Conversely, the two inclusions together say $\forall x\, (x \in A \Leftrightarrow x \in B)$ (Theorem 1.2.4), so $A = B$ by Axiom 2.2.1. ■

Part (3) is *the* standard way to prove two sets equal: show each is contained in the other.

2.3 The empty set

Axiom 2.3.1 (Empty set)**.** There is a set with no elements: $\exists E\ \forall x\ x \notin E$.

Theorem 2.3.2. *There is exactly one set with no elements.*

Proof. Existence is Axiom 2.3.1. If E and E' both have no elements, then $x \in E \Leftrightarrow x \in E'$ holds for every x, since both sides are false. So $E = E'$ by Axiom 2.2.1. ■

Definition 2.3.3 (Empty set)**.** This unique set is the *empty set*, written $\varnothing$.

Corollary 2.3.4. $\varnothing \subseteq A$ *for every set* A.

Proof. $x \in \varnothing \Rightarrow x \in A$ is true for every x, because its hypothesis is always false (Remark 1.2.2). ■

2.4 Forming subsets

Axiom 2.4.1 (Separation)**.** For every set A and every predicate $P(x)$ there is a set whose elements are exactly the elements of A that satisfy P:

$$\exists B\ \forall x\ (x \in B \Leftrightarrow (x \in A \wedge P(x))).$$

By Axiom 2.2.1, B is unique. We write it $\{x \in A \mid P(x)\}$.

This is really one axiom for each predicate P: an *axiom schema*. The predicate may mention other sets as parameters.

Separation only carves subsets out of a set we *already have*. That restriction is essential:

Theorem 2.4.2 (Russell)**.** *There is no set that contains every set.*

Proof. Suppose V were such a set. By Axiom 2.4.1, form $R = \{x \in V \mid x \notin x\}$. Since R is a set, $R \in V$. Then

$$R \in R \ \Leftrightarrow\ (R \in V \wedge R \notin R) \ \Leftrightarrow\ R \notin R,$$

a proposition equivalent to its own negation. That is a contradiction. ■

So "the collection of all sets" is not a set. Collections like that are sometimes called *classes*. We only use them as a way of speaking, never as objects.

2.5 Building larger sets

Axiom 2.5.1 (Pairing)**.** For all a, b there is a set whose elements are exactly a and b. By Axiom 2.2.1 it is unique, written $\{a, b\}$. We write $\{a\} \coloneqq \{a, a\}$, the *singleton* of a.

So $x \in \{a, b\} \Leftrightarrow (x = a \vee x = b)$, and $x \in \{a\} \Leftrightarrow x = a$.

Axiom 2.5.2 (Union)**.** For every set $\mathcal{F}$ there is a set whose elements are exactly the elements of the elements of $\mathcal{F}$. It is unique, written

$$\bigcup \mathcal{F}, \qquad x \in \bigcup \mathcal{F} \ \Leftrightarrow\ \exists F \in \mathcal{F}\ x \in F.$$

Definition 2.5.3 (Union, intersection, difference)**.** For sets A, B:

$$A \cup B \coloneqq \bigcup\{A, B\}, \qquad A \cap B \coloneqq \{x \in A \mid x \in B\}, \qquad A \setminus B \coloneqq \{x \in A \mid x \notin B\}.$$

For a *nonempty* set $\mathcal{F}$, pick any $F_0 \in \mathcal{F}$ and let

$$\bigcap \mathcal{F} \coloneqq \{x \in F_0 \mid \forall F \in \mathcal{F}\ x \in F\}.$$

This doesn't depend on the choice of F_0: any element of every $F \in \mathcal{F}$ is in particular in F_0. Sets A, B are *disjoint* if $A \cap B = \varnothing$. For three elements, $\{a, b, c\} \coloneqq \{a, b\} \cup \{c\}$, and so on.

Unwinding the definitions: $x \in A \cup B \Leftrightarrow x \in A \vee x \in B$; $x \in A \cap B \Leftrightarrow x \in A \wedge x \in B$; $x \in A \setminus B \Leftrightarrow x \in A \wedge x \notin B$.

Remark 2.5.4. $\bigcap \varnothing$ is left undefined on purpose. "The set of all x in every $F \in \varnothing$" would contain every set, which Theorem 2.4.2 forbids.

Theorem 2.5.5 (Algebra of sets)**.** *For all sets A, B, C:*

Law	*Statement*
Commutativity	$A \cup B = B \cup A, \quad A \cap B = B \cap A$
Associativity	$(A \cup B) \cup C = A \cup (B \cup C), \quad$ *same for* $\cap$
Distributivity	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
De Morgan	$C \setminus (A \cup B) = (C \setminus A) \cap (C \setminus B)$
	$C \setminus (A \cap B) = (C \setminus A) \cup (C \setminus B)$
Absorption	$A \cup (A \cap B) = A, \quad A \cap (A \cup B) = A$
Inclusion	$A \subseteq B \Leftrightarrow A \cup B = B \Leftrightarrow A \cap B = A$

Proof. Each law turns into the matching law of logic (Theorem 1.2.4) once we unwind membership. For example, for the first distributive law and any x:

$$\begin{aligned} x \in A \cap (B \cup C) &\Leftrightarrow x \in A \wedge (x \in B \vee x \in C) \\ &\Leftrightarrow (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) \\ &\Leftrightarrow x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

So the two sets have the same elements and are equal by Axiom 2.2.1. For the first De Morgan law: $x \in C \wedge \neg(x \in A \vee x \in B) \Leftrightarrow (x \in C \wedge x \notin A) \wedge (x \in C \wedge x \notin B)$. The remaining laws work the same way. ∎

Axiom 2.5.6 (Power set)**.** For every set A there is a set whose elements are exactly the subsets of A. It is unique, written $\mathcal{P}A$:

$$X \in \mathcal{P}A \ \Leftrightarrow\ X \subseteq A.$$

For example, $\mathcal{P}\varnothing = \{\varnothing\}$ and $\mathcal{P}\{\varnothing\} = \{\varnothing, \{\varnothing\}\}$. Note $\varnothing \neq \{\varnothing\}$: the first has no elements, the second has one.

So from nothing, the empty set, we already get infinitely many different sets: $\varnothing$, $\{\varnothing\}$, $\{\{\varnothing\}\}$, $\{\varnothing, \{\varnothing\}\}$, … In Chapter 5 some of them become the natural numbers.

2.6 The remaining axioms

ZF has four more axioms:

- **Infinity** guarantees a set containing all natural numbers. It is introduced in Chapter 5, where it is needed.
- **Replacement** says the image of a set under a definable rule is a set. Nothing in the chapters on numbers needs it. It will be introduced when it is.
- **Foundation** says every nonempty set A has an element disjoint from A. It rules out sets like $x \in x$. We won't need it: Lemma 5.2.2 proves $n \notin n$ for natural numbers directly.
- **Choice** (which turns ZF into ZFC) is not used in the number constructions. Whenever it is used later, that will be stated.

Exercises

1. Prove $A \setminus (A \setminus B) = A \cap B$.
2. Prove $A \subseteq B \Rightarrow \mathcal{P}A \subseteq \mathcal{P}B$.
3. Write out the elements of $\mathcal{P}\{\varnothing, \{\varnothing\}\}$. How many are there?
4. Show that for every set A there is a set not in A. (Hint: adapt the proof of Theorem 2.4.2.)

Chapter 3

Relations

Sets don't care about order: $\{a, b\} = \{b, a\}$. To compare, combine or map objects we need a pair in which one element comes *first*. Then we use pairs to say when objects are *related*, and from that we get the two most important kinds of relations: *equivalences* and *orders*.

3.1 Ordered pairs

Definition 3.1.1 (Ordered pair, Kuratowski)**.**

$$(a, b) \coloneqq \{\{a\}, \{a, b\}\}.$$

a is the *first component* and b the *second component*.

This exists by Axiom 2.5.1. The definition is only useful because of the following property, which is all we will ever use about pairs.

Theorem 3.1.2 (Characteristic property of pairs)**.**

$$(a, b) = (c, d) \;\Leftrightarrow\; a = c \wedge b = d.$$

Proof. ($\Leftarrow$) Substitution.

($\Rightarrow$) Assume $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$.

Step 1: $a = c$. $\{a\}$ is an element of the left side, so of the right side: $\{a\} = \{c\}$ or $\{a\} = \{c, d\}$. In both cases $c \in \{a\}$, so $c = a$.

Step 2, case $a = b$. Then $(a, b) = \{\{a\}, \{a\}\} = \{\{a\}\}$. The element $\{c, d\}$ of the right side must equal $\{a\}$, so $d \in \{a\}$, so $d = a = b$.

Step 2, case $a \neq b$. $\{a, b\}$ is an element of the right side $\{\{a\}, \{a, d\}\}$ (using $c = a$). It is not $\{a\}$, since $b \in \{a, b\}$ but $b \notin \{a\}$. So $\{a, b\} = \{a, d\}$. Then $b \in \{a, d\}$ and $b \neq a$, so $b = d$. ∎

Definition 3.1.3 (Tuples)**.** Let $(a, b, c) \coloneqq ((a, b), c)$, and generally

$$(a_1, \ldots, a_{n+1}) \coloneqq ((a_1, \ldots, a_n), a_{n+1}).$$

Applying Theorem 3.1.2 repeatedly, two tuples are equal if and only if their components are equal one by one.

3.2 Cartesian products

Theorem 3.2.1. *For sets A, B there is a unique set whose elements are exactly the pairs (a, b) with $a \in A$ and $b \in B$.*

Proof. We need a set we already have that contains all such pairs, so that Separation can pick them out. Let $a \in A$ and $b \in B$. Then $\{a\} \subseteq A \cup B$ and $\{a, b\} \subseteq A \cup B$, so both are elements of $\mathcal{P}(A \cup B)$. Hence $(a, b) \subseteq \mathcal{P}(A \cup B)$, so $(a, b) \in \mathcal{PP}(A \cup B)$. The set

$$\{p \in \mathcal{PP}(A \cup B) \mid \exists a \in A\ \exists b \in B\ p = (a, b)\}$$

has the required elements. Uniqueness is by Axiom 2.2.1. ■

Definition 3.2.2 (Cartesian product)**.** The set from Theorem 3.2.1 is the *Cartesian product* $A \times B$. For a predicate $P(x, y)$ we write

$$\{(x, y) \in A \times B \mid P(x, y)\} \coloneqq \{p \in A \times B \mid \exists x \in A\ \exists y \in B\ (p = (x, y) \wedge P(x, y))\}.$$

3.3 Relations

Definition 3.3.1 (Relation)**.** A *relation from A to B* is a subset $R \subseteq A \times B$. A *relation on* A is a relation from A to A. For $(x, y) \in R$ we write

$$x\ R\ y,$$

and by Rule 2 the form $(x, y) \in R$ is not used again. The *domain*, *range* and *inverse* of R are

$$\operatorname{dom} R \coloneqq \{x \in A \mid \exists y \in B\ x\ R\ y\}, \qquad \operatorname{ran} R \coloneqq \{y \in B \mid \exists x \in A\ x\ R\ y\},$$
$$R^{-1} \coloneqq \{(y, x) \in B \times A \mid x\ R\ y\}.$$

So R^{-1} is a relation from B to A.

Definition 3.3.2 (Identity relation)**.** The *identity* on A is $\mathrm{id}_A \coloneqq \{(x, y) \in A \times A \mid x = y\}$.

Other examples on a set A: the empty relation $\varnothing$ and the full relation $A \times A$. On $\mathcal{P}A$ the statement $X \subseteq Y$ is itself a relation, the set $\{(X, Y) \in \mathcal{P}A \times \mathcal{P}A \mid X \subseteq Y\}$, and Rule 5 lets $\subseteq$ name both.

Definition 3.3.3 (Properties of relations)**.** A relation R on A is

- *reflexive* if $\forall x \in A\ x\ R\ x$;
- *symmetric* if $\forall x, y \in A\ (x\ R\ y \Rightarrow y\ R\ x)$;
- *antisymmetric* if $\forall x, y \in A\ (x\ R\ y \wedge y\ R\ x \Rightarrow x = y)$;
- *transitive* if $\forall x, y, z \in A\ (x\ R\ y \wedge y\ R\ z \Rightarrow x\ R\ z)$;
- *irreflexive* if $\forall x \in A\ \neg(x\ R\ x)$.

3.4 Equivalence relations

An equivalence relation says when two objects are "the same in the respect we care about". It is the main tool of the number constructions: integers and rationals are both *equivalence classes*.

Definition 3.4.1 (Equivalence relation)**.** An *equivalence relation* on A is a relation $\sim$ on A that is reflexive, symmetric and transitive. For $a \in A$, the *equivalence class* of a is

$$[a] \coloneqq \{x \in A \mid x \sim a\}.$$

The *quotient set* is the set of all classes:

$$A/\!\sim \coloneqq \{C \in \mathcal{P}A \mid \exists a \in A\ C = [a]\}.$$

Any $x \in C$ is a *representative* of the class C.

Lemma 3.4.2. *Let $\sim$ be an equivalence relation on A and $a, b \in A$.*

(1) $a \in [a]$.

(2) $[a] = [b] \ \Leftrightarrow\ a \sim b$.

(3) Either $[a] = [b]$ *or* $[a] \cap [b] = \varnothing$.

Proof. (1) Reflexivity.

(2) ($\Rightarrow$) $a \in [a] = [b]$, so $a \sim b$. ($\Leftarrow$) Let $x \in [a]$, so $x \sim a$. With $a \sim b$ and transitivity, $x \sim b$, so $x \in [b]$. So $[a] \subseteq [b]$. By symmetry $b \sim a$, and the same argument gives $[b] \subseteq [a]$. Use Theorem 2.2.3.

(3) Suppose some x lies in both $[a]$ and $[b]$. Then $x \sim a$ and $x \sim b$, so $a \sim x$ (symmetry) and $a \sim b$ (transitivity). By (2), $[a] = [b]$. ■

So every element sits in exactly one class, and *which* representative we use to name a class doesn't matter.

Definition 3.4.3 (Partition)**.** A *partition* of A is a set Π of nonempty subsets of A such that every $x \in A$ lies in exactly one $C \in \Pi$.

Theorem 3.4.4 (Equivalences are partitions)**.**

(1) If $\sim$ is an equivalence relation on A, then $A/\!\sim$ is a partition of A.

(2) If Π is a partition of A, then the relation $\approx$ on A given by $x \approx y :\Leftrightarrow \exists C \in \Pi\ (x \in C \wedge y \in C)$ is an equivalence relation whose classes are exactly the elements of Π.

Proof. (1) Every class $[a]$ is nonempty, since it contains a (Lemma 3.4.2(1)). Every $x \in A$ is in the class $[x]$, and if $x \in [a]$ and $x \in [b]$, then $[a] = [b]$ by Lemma 3.4.2(3).

(2) Reflexive: x lies in some $C \in \Pi$. Symmetric: the definition is symmetric in x, y. Transitive: if $x, y \in C$ and $y, z \in C'$, then y is in both C and C', so $C = C'$ by uniqueness, and $x, z \in C$. For the classes: let C be the element of Π containing a. Then $x \approx a$ if and only if x and a share an element of Π, if and only if $x \in C$ (uniqueness again). So $[a] = C$, and every $C \in \Pi$ arises this way because $C \neq \varnothing$. ■

Definition 3.4.5 (Well-defined)**.** Often we define something on classes by using a representative: "for a class C, pick $a \in C$ and let the result be $\ldots\ a\ \ldots$". This is a valid definition only if the result is the same for *every* choice of representative. Proving that is called *showing that the definition is well-defined*.

We will do this many times in Chapters 6 and 7.

3.5 Orders

Definition 3.5.1 (Partial and total order)**.** A relation $\leq$ on A is a *partial order* if it is reflexive, antisymmetric and transitive. Then $(A, \leq)$ is a *partially ordered set*. It is a *total order* if in addition any two elements are comparable:

$$\forall x, y \in A\ (x \leq y \vee y \leq x).$$

We write $x < y$ for $x \leq y \wedge x \neq y$. By Rule 5 there is no separate symbol for the reversed relations.

For example, $\subseteq$ on $\mathcal{P}A$ is a partial order by Theorem 2.2.3. It is not total in general: on $\mathcal{P}\{a, b\}$ with $a \neq b$, neither $\{a\} \subseteq \{b\}$ nor $\{b\} \subseteq \{a\}$.

Lemma 3.5.2. *If $\leq$ is a partial order, then $<$ is irreflexive and transitive. Conversely, if $<$ is an irreflexive transitive relation on A, then $x \leq y :\Leftrightarrow x < y \vee x = y$ is a partial order.*

Proof. Irreflexivity of $<$ is built into its definition. Transitivity: if $x < y$ and $y < z$, then $x \leq z$. If $x = z$, then $x \leq y$ and $y \leq x$, so $x = y$ by antisymmetry, contradicting $x < y$. So $x < z$.

Conversely, $\leq$ is reflexive by definition. Transitivity follows by cases on which side of each "or" holds. Antisymmetry: if $x \leq y$, $y \leq x$ and $x \neq y$, then $x < y$ and $y < x$, so $x < x$ by transitivity, contradicting irreflexivity. ■

Definition 3.5.3 (Bounds)**.** Let $(A, \leq)$ be partially ordered and $S \subseteq A$.

- $m \in S$ is a *least element* of S if $m \leq s$ for all $s \in S$, and a *greatest element* if $s \leq m$ for all $s \in S$.
- $u \in A$ is an *upper bound* of S if $s \leq u$ for all $s \in S$, and a *lower bound* if $u \leq s$ for all $s \in S$. S is *bounded above* (*below*) if it has an upper (lower) bound.
- A least element of the set of upper bounds of S is a *supremum* $\sup S$. A greatest element of the set of lower bounds is an *infimum* $\inf S$.

Lemma 3.5.4. *A set has at most one least element. So $\sup S$ and $\inf S$ are unique when they exist.*

Proof. If m and m' are least elements of S, then $m \leq m'$ and $m' \leq m$, so $m = m'$ by antisymmetry. The same works for greatest elements. ■

Suprema become central with the real numbers: $\mathbb{R}$ is constructed so that *every* nonempty set bounded above has a supremum, which $\mathbb{Q}$ lacks (Theorem 7.6.1).

Definition 3.5.5 (Well-order)**.** A total order $\leq$ on A is a *well-order* if every nonempty subset of A has a least element.

Exercises

1. Show that $(a, b, c) = (a', b', c')$ if and only if $a = a'$, $b = b'$ and $c = c'$.
2. Show $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
3. On a set A, is the empty relation reflexive? Symmetric? Transitive? (Watch out when $A = \varnothing$.)
4. Find an example showing that "symmetric and transitive" does not imply reflexive.
5. In $(\mathcal{P}A, \subseteq)$, show that every $\mathcal{F} \subseteq \mathcal{P}A$ has supremum $\bigcup \mathcal{F}$.

Chapter 4

Maps

A map assigns to each element of one set exactly one element of another. Maps are needed *before* numbers: the successor of a number, and addition and multiplication, are all maps. Later, the *functions* of calculus will be special maps between sets of real numbers. To keep the two apart, this book says *map* for the general notion.

This chapter also fixes the one notation for maps that the whole book uses (Rule 3).

4.1 Maps as relations

Definition 4.1.1 (Map)**.** A *map* f from A to B, written $f\colon A \to B$, is a relation $f \subseteq A \times B$ that relates every element of A to exactly one element of B. For a map, the relation $x\ f\ y$ is written

$$x \overset{f}{\longmapsto} y \qquad (\text{“}f \text{ sends } x \text{ to } y\text{”}),$$

so the defining condition reads $\forall x \in A\ \exists! y \in B\ x \overset{f}{\longmapsto} y$. That unique y is called the *value* of f at x. By Rule 3 a value is only ever named in words or by a letter introduced for it, never by a term such as “f of x”. A is the *domain* of f and B its *codomain*.

So a map *is* its graph: the set of all pairs (x, y) with $x \overset{f}{\longmapsto} y$. A map with domain $\mathbb{N}$ (Chapter 5) is called a *sequence*, and a map with domain I a *family indexed by* I. These are only names: sequences and families are maps and use the same notation.

Theorem 4.1.2 (Equality of maps)**.** *For maps $f, g\colon A \to B$:*

$$f = g \ \Leftrightarrow\ \forall x \in A\ \forall y \in B\ (x \overset{f}{\longmapsto} y \Rightarrow x \overset{g}{\longmapsto} y).$$

Proof. ($\Rightarrow$) Substitution. ($\Leftarrow$) Let $p \in f$. Since $f \subseteq A \times B$, $p = (x, y)$ with $x \overset{f}{\longmapsto} y$. Then $x \overset{g}{\longmapsto} y$, so $p \in g$. So $f \subseteq g$. Conversely, let $p = (x, y) \in g$, and let y' be the value of f at x. Then $x \overset{g}{\longmapsto} y'$ by hypothesis, so $y' = y$ because g has only one value at x. So $p \in f$. Use Theorem 2.2.3. ∎

Theorem 4.1.3 (Defining maps by a rule)**.** *Let $P(x, y)$ be a predicate such that for every $x \in A$ there is exactly one $y \in B$ with $P(x, y)$. Then there is exactly one map $f\colon A \to B$ with*

$$\forall x \in A\ \forall y \in B\ \left(x \overset{f}{\longmapsto} y \Leftrightarrow P(x, y)\right).$$

Proof. $f \coloneqq \{(x, y) \in A \times B \mid P(x, y)\}$ exists by Definition 3.2.2 and is a map by the hypothesis. Uniqueness follows from Theorem 4.1.2. ∎

The most common case is $P(x, y) :\Leftrightarrow y = t$ for an expression t built from x that lies in B for every $x \in A$. We then speak of *the map* $f\colon A \to B$ *with* $x \overset{f}{\longmapsto} t$ *for all* $x \in A$.

Definition 4.1.4 (Set of maps)**.** $B^A \coloneqq \{f \in \mathcal{P}(A \times B) \mid f \text{ is a map from } A \text{ to } B\}$.

For example, id_A (Definition 3.3.2) is the map $A \to A$ with $x \overset{\mathrm{id}_A}{\longmapsto} x$. For any A there is exactly one map $\varnothing \to A$, namely the empty relation: the condition "for every $x \in \varnothing$" is vacuously true.

4.2 Images and preimages

Definition 4.2.1 (Image, preimage, restriction)**.** Let R be a relation from A to B and $X \subseteq A$. The *image* of X under R is

$$R[X] \coloneqq \{y \in B \mid \exists x \in X\ x\ R\ y\}.$$

For a map $f\colon A \to B$ and $Y \subseteq B$ this gives the image $f[X] = \{y \in B \mid \exists x \in X\ x \overset{f}{\longmapsto} y\}$ and the *preimage*

$$f^{-1}[Y] = \{x \in A \mid \exists y \in Y\ x \overset{f}{\longmapsto} y\},$$

the image of Y under the inverse relation f^{-1}. The *restriction* of f to X is $f|_X \coloneqq f \cap (X \times B)$, a map $X \to B$ with $x \overset{f|_X}{\longmapsto} y \Leftrightarrow x \overset{f}{\longmapsto} y$ for $x \in X$.

The image of a *set* is written with square brackets (Rule 6). The value of a map at an *element* has no bracket notation at all, so the two can never be confused.

4.3 Injective, surjective, bijective

Definition 4.3.1. A map $f\colon A \to B$ is

- *injective* if different elements have different values: $\forall x, x' \in A\ \forall y \in B\ (x \overset{f}{\longmapsto} y \wedge x' \overset{f}{\longmapsto} y \Rightarrow x = x')$;
- *surjective* if every element of B is a value: $\forall y \in B\ \exists x \in A\ x \overset{f}{\longmapsto} y$, that is, $f[A] = B$;
- *bijective* if it is both. Then every $y \in B$ is the value of exactly one $x \in A$.

4.4 Composition and inverses

Definition 4.4.1 (Composition)**.** For a relation R from A to B and a relation S from B to C, the *composition* is the relation from A to C

$$S \circ R \coloneqq \{(x, z) \in A \times C \mid \exists y \in B\ (x\ R\ y \wedge y\ S\ z)\}.$$

Theorem 4.4.2 (Composition of maps)**.** *For maps* $f\colon A \to B$ *and* $g\colon B \to C$*, the composition* $g \circ f$ *is a map* $A \to C$*, and*

$$x \overset{f}{\longmapsto} y\ \wedge\ y \overset{g}{\longmapsto} z\ \Rightarrow\ x \overset{g \circ f}{\longmapsto} z.$$

Proof. Let $x \in A$, let y be the value of f at x, and z the value of g at y. Then $x\ (g \circ f)\ z$ by definition. If also $x\ (g \circ f)\ z'$, there is y' with $x \overset{f}{\longmapsto} y'$ and $y' \overset{g}{\longmapsto} z'$. Then $y' = y$, since f has one value at x, and then $z' = z$, since g has one value at y. ∎

Definition 4.4.3 (Chains)**.** A chain $x \overset{f}{\longmapsto} y \overset{g}{\longmapsto} z$ abbreviates $x \overset{f}{\longmapsto} y \wedge y \overset{g}{\longmapsto} z$, and likewise for longer chains.

Theorem 4.4.4. *For $f\colon A \to B$, $g\colon B \to C$ and $h\colon C \to D$:*

(1) $h \circ (g \circ f) = (h \circ g) \circ f$ (associativity);

(2) $f \circ \mathrm{id}_A = f = \mathrm{id}_B \circ f$;

(3) if f and g are injective (surjective, bijective), then so is $g \circ f$.

Proof. (1) Both sides send x to w exactly when there are y, z with $x \overset{f}{\longmapsto} y$, $y \overset{g}{\longmapsto} z$ and $z \overset{h}{\longmapsto} w$. Use Theorem 4.1.2.

(2) $x \overset{\mathrm{id}_A}{\longmapsto} x \overset{f}{\longmapsto} y$ and $x \overset{f}{\longmapsto} y \overset{\mathrm{id}_B}{\longmapsto} y$, so all three maps send x to the value of f at x.

(3) Injective: let $x \overset{f}{\longmapsto} y \overset{g}{\longmapsto} z$ and $x' \overset{f}{\longmapsto} y' \overset{g}{\longmapsto} z$. Then $y = y'$ because g is injective, and then $x = x'$ because f is injective. Surjective: for $z \in C$ pick $y \in B$ with $y \overset{g}{\longmapsto} z$, then $x \in A$ with $x \overset{f}{\longmapsto} y$. Then $x \overset{g\circ f}{\longmapsto} z$. Bijective: combine both. ∎

Theorem 4.4.5 (Inverse map)**.** *A map $f\colon A \to B$ is bijective if and only if there is a map $g\colon B \to A$ with*

$$g \circ f = \mathrm{id}_A \quad \textit{and} \quad f \circ g = \mathrm{id}_B.$$

In that case g is unique and equals the inverse relation f^{-1} (Definition 3.3.1), which is therefore a map, the inverse *of f:*

$$y \overset{f^{-1}}{\longmapsto} x \;\Leftrightarrow\; x \overset{f}{\longmapsto} y.$$

Proof. ($\Rightarrow$) Since f is bijective, every $y \in B$ is the value of exactly one $x \in A$, so the inverse relation f^{-1} is a map $B \to A$. If $x \overset{f}{\longmapsto} y \overset{f^{-1}}{\longmapsto} x'$, then $x' \overset{f}{\longmapsto} y$, so $x' = x$ by injectivity: $f^{-1} \circ f$ sends every x to x. If $y \overset{f^{-1}}{\longmapsto} x \overset{f}{\longmapsto} y'$, then $x \overset{f}{\longmapsto} y$, so $y' = y$: $f \circ f^{-1}$ sends every y to y. By Theorem 4.1.2, $g \coloneqq f^{-1}$ works.

($\Leftarrow$) Injective: let $x \overset{f}{\longmapsto} y$ and $x' \overset{f}{\longmapsto} y$, and let z be the value of g at y. Then $x \overset{g\circ f}{\longmapsto} z$ and $x' \overset{g\circ f}{\longmapsto} z$ (Theorem 4.4.2). Since $g \circ f = \mathrm{id}_A$, $z = x$ and $z = x'$. Surjective: let $y \in B$. Since $f \circ g = \mathrm{id}_B$, $y \overset{f\circ g}{\longmapsto} y$, so there is x with $y \overset{g}{\longmapsto} x \overset{f}{\longmapsto} y$.

Uniqueness: if g' also works, then by Theorem 4.4.4

$$g' = g' \circ \mathrm{id}_B = g' \circ (f \circ g) = (g' \circ f) \circ g = \mathrm{id}_A \circ g = g.$$

Since f^{-1} works, every such g equals f^{-1}. ∎

4.5 Maps on quotient sets

When we construct $\mathbb{Z}$ and $\mathbb{Q}$, arithmetic is first defined on representatives and then moved to classes. The following theorem makes that precise.

Theorem 4.5.1 (Inducing a map on a quotient)**.** *Let $\sim$ be an equivalence relation on A and $f\colon A \to B$ a map that respects $\sim$:*

$$a \sim a' \wedge a \overset{f}{\longmapsto} b \wedge a' \overset{f}{\longmapsto} b' \;\Rightarrow\; b = b'.$$

Then there is exactly one map $\bar{f}\colon A/{\sim} \to B$ with

$$a \overset{f}{\longmapsto} b \;\Rightarrow\; [a] \overset{\bar{f}}{\longmapsto} b \qquad \textit{for all } a \in A,\ b \in B.$$

Proof. For a class $C \in A/\sim$ we need exactly one b with $a \overset{f}{\longmapsto} b$ for some $a \in C$. There is at least one, because C is nonempty (Theorem 3.4.4). There is at most one: if $a, a' \in C$, then $a \sim a'$ (Lemma 3.4.2), so their values agree. Now apply Theorem 4.1.3. Uniqueness follows from Theorem 4.1.2, because every class is of the form $[a]$. ■

The hypothesis "f respects $\sim$" is exactly the statement that $\bar{f}$ is well-defined (Definition 3.4.5).

4.6 Operations

Definition 4.6.1 (Operation)**.** A *binary operation* on a set A is a map $*\colon A \times A \to A$ that has been given a symbol of its own. Its value is written as a term (Rule 4): $a * b$ denotes the value of $*$ at (a, b), so

$$(a, b) \overset{*}{\longmapsto} c \;\Leftrightarrow\; a * b = c.$$

Likewise a *unary operation* is a map $A \to A$ with a symbol of its own, whose value at a is written with that symbol, as in $-a$. An operation $*$ is *associative* if $(a * b) * c = a * (b * c)$ for all a, b, c, and *commutative* if $a * b = b * a$ for all a, b. An element $e \in A$ is *neutral* for $*$ if $e * a = a = a * e$ for all a.

Constructions that apply to all sets, such as $\cup$ and $\circ$, are written in the same way. A neutral element is unique when it exists: if e and e' are both neutral, then $e = e * e' = e'$.

Exercises

1. Let $f\colon A \to B$ and $X, X' \subseteq A$. Show $f[X \cup X'] = f[X] \cup f[X']$ and $f[X \cap X'] \subseteq f[X] \cap f[X']$. Find an example where the inclusion is strict. Show it is an equality when f is injective.

2. Show that preimages preserve everything: $f^{-1}[Y \cap Y'] = f^{-1}[Y] \cap f^{-1}[Y']$, and the same for $\cup$ and $\setminus$.

3. If $g \circ f$ is injective, must f be injective? Must g be?

4. Show: for $A \neq \varnothing$, $f\colon A \to B$ is injective if and only if there is a $g\colon B \to A$ with $g \circ f = \mathrm{id}_A$.

Chapter 5

Natural numbers

We now construct the numbers $0, 1, 2, 3, \ldots$ out of sets. The idea, due to von Neumann, is to let *each number be the set of all numbers before it*:

$$0 = \varnothing, \quad 1 = \{0\}, \quad 2 = \{0, 1\}, \quad 3 = \{0, 1, 2\}, \quad \ldots$$

Then the number n has exactly n elements, "m is smaller than n" can simply mean $m \in n$, and passing to the next number means adding the number itself as a new element.

Then we prove that these sets behave the way numbers should. The *Peano axioms*, induction included, come out as *theorems*. So do addition, multiplication and order, with all the usual laws.

5.1 Construction

Definition 5.1.1 (Successor, inductive set)**.** The *successor* of a set x is the set $x \cup \{x\}$. A set I is *inductive* if $\varnothing \in I$ and $\forall x\ (x \in I \Rightarrow x \cup \{x\} \in I)$.

An inductive set contains $\varnothing$, $\varnothing \cup \{\varnothing\}$, the successor of that, and so on. None of the axioms so far guarantees that such a set exists: every set we have built so far is finite. So we need a new axiom.

Axiom 5.1.2 (Infinity)**.** There is an inductive set.

Definition 5.1.3 (Natural numbers)**.** Let I_0 be an inductive set (Axiom 5.1.2) and

$$\mathbb{N} \coloneqq \{x \in I_0 \mid x \in I \text{ for every inductive set } I\}.$$

The elements of $\mathbb{N}$ are the *natural numbers*. The *numerals* are

$$0 \coloneqq \varnothing, \quad 1 \coloneqq 0 \cup \{0\}, \quad 2 \coloneqq 1 \cup \{1\}, \quad \ldots, \quad 9 \coloneqq 8 \cup \{8\}.$$

$\mathbb{N}$ doesn't depend on which I_0 we picked: if I_1 is another inductive set, any x in every inductive set is in both I_0 and I_1, so both choices give the same elements. Two facts follow directly from the definition:

- $\mathbb{N}$ *is inductive.* $\varnothing$ lies in every inductive set. If x lies in every inductive set, so does $x \cup \{x\}$.
- $\mathbb{N}$ *is contained in every inductive set.* That is how its elements were chosen.

So $\mathbb{N}$ is the *smallest* inductive set. Indeed $1 = \varnothing \cup \{\varnothing\} = \{0\}$, $2 = \{0\} \cup \{1\} = \{0, 1\}$, and so on. Positional notation like $10, 11, \ldots$ needs division with remainder and waits for the chapter on divisibility. In this book, $0 \in \mathbb{N}$.

5.2 Induction

Theorem 5.2.1 (Principle of induction)**.** *Let $X \subseteq \mathbb{N}$ with*

(1) $0 \in X$, and

(2) $n \in X \Rightarrow n \cup \{n\} \in X$ for all $n \in \mathbb{N}$.

Then $X = \mathbb{N}$.

Equivalently, for a predicate P: if $P(0)$ holds (the base case*) and $P(n) \Rightarrow P(n \cup \{n\})$ holds for every $n \in \mathbb{N}$ (the* induction step*, with* induction hypothesis *$P(n)$), then $P(n)$ holds for all $n \in \mathbb{N}$.*

Proof. By (1) and (2), X is inductive: every element of X lies in $\mathbb{N}$, so (2) applies to it. So $\mathbb{N} \subseteq X$ (Definition 5.1.3), and $X \subseteq \mathbb{N}$ is given. For the predicate form, apply this to $X \coloneqq \{n \in \mathbb{N} \mid P(n)\}$. ■

Induction is not a new law of logic. It follows from the way $\mathbb{N}$ was built: as the *smallest* inductive set, it has no room for anything that isn't reached from 0 by successors.

Lemma 5.2.2 (Shape of natural numbers)**.** *For every $n \in \mathbb{N}$:*

(1) $n \subseteq \mathbb{N}$: every element of a natural number is a natural number;

(2) n is transitive*: $m \in n \Rightarrow m \subseteq n$;*

(3) $n \notin n$.

Proof. Each by induction on n.

(1) $0 = \varnothing \subseteq \mathbb{N}$. If $n \subseteq \mathbb{N}$, then $n \cup \{n\} \subseteq \mathbb{N}$, because $n \in \mathbb{N}$.

(2) 0 has no elements, so the condition holds vacuously. Assume n is transitive and let $m \in n \cup \{n\}$. Then $m \in n$ or $m = n$. If $m \in n$, then $m \subseteq n \subseteq n \cup \{n\}$. If $m = n$, then $m = n \subseteq n \cup \{n\}$.

(3) $0 \notin 0$, since 0 has no elements. Assume $n \notin n$, and suppose $n \cup \{n\} \in n \cup \{n\}$. Then $n \cup \{n\} \in n$ or $n \cup \{n\} = n$. If $n \cup \{n\} = n$, then $n \in n \cup \{n\} = n$. If $n \cup \{n\} \in n$, then $n \cup \{n\} \subseteq n$ by (2), and $n \in n \cup \{n\}$ gives $n \in n$. Both contradict the induction hypothesis. ■

Definition 5.2.3 (Successor map)**.** Since $\mathbb{N}$ is inductive, $n \cup \{n\} \in \mathbb{N}$ for every $n \in \mathbb{N}$. The *successor map* is the map $S \colon \mathbb{N} \to \mathbb{N}$ with $n \overset{S}{\longmapsto} n \cup \{n\}$.

Theorem 5.2.4 (Peano axioms)**.**

(1) $0 \in \mathbb{N}$.

(2) S is a map $\mathbb{N} \to \mathbb{N}$.

(3) 0 is not a value of S: $n \cup \{n\} \neq 0$ for every $n \in \mathbb{N}$.

(4) S is injective: $m \cup \{m\} = n \cup \{n\} \Rightarrow m = n$.

(5) Induction (Theorem 5.2.1).

Proof. (1) and (2): $\mathbb{N}$ is inductive. (3): $n \in n \cup \{n\}$, so $n \cup \{n\} \neq \varnothing$.

(4) Let $m \cup \{m\} = n \cup \{n\}$ and suppose $m \neq n$. Since m is an element of the left side, $m \in n$ or $m = n$, so $m \in n$. In the same way $n \in m$. By transitivity (Lemma 5.2.2(2)), $m \in n$ gives $m \subseteq n$, and with $n \in m$ that gives $n \in n$, contradicting Lemma 5.2.2(3). ■

Corollary 5.2.5 (Predecessor)**.** *For every $n \in \mathbb{N}$ with $n \neq 0$ there is exactly one $p \in \mathbb{N}$ with $n = p \cup \{p\}$, the* predecessor *of n.*

Proof. Existence, by induction on the statement "$n = 0$ or $n = p \cup \{p\}$ for some $p \in \mathbb{N}$": it holds for 0, and for $n \cup \{n\}$ with $p := n$. Uniqueness: Theorem 5.2.4(4). ■

5.3 Recursion

Induction *proves* statements about all n. Recursion *defines* maps on $\mathbb{N}$: give the value at 0, and a rule that turns the value at n into the value at the successor of n. That this defines a map at all is a theorem.

Theorem 5.3.1 (Recursion theorem)**.** *Let A be a set, $a \in A$, and $g\colon A \to A$ a map. Then there is exactly one map $h\colon \mathbb{N} \to A$ with*

$$0 \overset{h}{\longmapsto} a \qquad \text{and} \qquad n \overset{h}{\longmapsto} y \overset{g}{\longmapsto} z \;\Rightarrow\; n \cup \{n\} \overset{h}{\longmapsto} z \quad \text{for all } n \in \mathbb{N},\ y, z \in A.$$

In terms of composition, the second condition says $h \circ S = g \circ h$.

Proof. Idea: h should be the smallest relation that relates 0 to a and is closed under the step from n to its successor. Then we show this relation is a map.

Call a relation $R \subseteq \mathbb{N} \times A$ *closed* if $0\ R\ a$ and $n\ R\ y \wedge y \overset{g}{\longmapsto} z \Rightarrow (n \cup \{n\})\ R\ z$. Then $\mathbb{N} \times A$ is closed. Let h be the intersection of all closed relations, i.e. of the nonempty set of closed elements of $\mathcal{P}(\mathbb{N} \times A)$ (Definition 2.5.3). Then h is closed as well, and $h \subseteq R$ for every closed R.

h is a map. We show by induction on n: there is exactly one y with $n\ h\ y$.

- $n = 0$. We have $0\ h\ a$. Suppose $0\ h\ b$ with $b \neq a$. Then $h' := h \setminus \{(0, b)\}$ is still closed: it relates 0 to a, and every pair added by the step has as first component a successor, which is not 0 (Theorem 5.2.4(3)). That contradicts $h \subseteq h'$, since $(0, b) \notin h'$.
- Step. Assume y is the only element with $n\ h\ y$, and let z be the value of g at y. Since h is closed, $(n \cup \{n\})\ h\ z$. Suppose $(n \cup \{n\})\ h\ w$ with $w \neq z$. Then $h' := h \setminus \{(n \cup \{n\}, w)\}$ is still closed. It relates 0 to a. If $k\ h'\ u \overset{g}{\longmapsto} v$ and $(k \cup \{k\}, v) = (n \cup \{n\}, w)$, then $k = n$ (Theorem 5.2.4(4)), so $u = y$ by the induction hypothesis, so $v = z$, which contradicts $v = w \neq z$. Again $h \subseteq h'$ fails.

So h is a map, and since it is closed, it has both properties.

Uniqueness. If h' also has both properties, we show by induction that h and h' have the same value at every n: both send 0 to a, and if both send n to y, and $y \overset{g}{\longmapsto} z$, both send $n \cup \{n\}$ to z. Use Theorem 4.1.2. ■

Often the step also needs to know n, as for the factorial. The stronger form follows from the simple one.

Corollary 5.3.2 (Recursion with index)**.** *Let $a \in A$ and $g\colon \mathbb{N} \times A \to A$. Then there is exactly one $h\colon \mathbb{N} \to A$ with*

$$0 \overset{h}{\longmapsto} a \qquad \text{and} \qquad n \overset{h}{\longmapsto} y \wedge (n, y) \overset{g}{\longmapsto} z \;\Rightarrow\; n \cup \{n\} \overset{h}{\longmapsto} z.$$

Proof. Let $G\colon \mathbb{N} \times A \to \mathbb{N} \times A$ be the map that sends (n, y) to $(n \cup \{n\}, z)$, where z is the value of g at (n, y) (Theorem 4.1.3). Theorem 5.3.1, applied to $\mathbb{N} \times A$, the start value $(0, a)$ and G, gives a map $H\colon \mathbb{N} \to \mathbb{N} \times A$. By induction, for every n there is a y with $n \overset{H}{\longmapsto} (n, y)$: this holds for 0, and if $n \overset{H}{\longmapsto} (n, y) \overset{G}{\longmapsto} (n \cup \{n\}, z)$, then $n \cup \{n\} \overset{H}{\longmapsto} (n \cup \{n\}, z)$. So

$$h := \{(n, y) \in \mathbb{N} \times A \mid n \overset{H}{\longmapsto} (n, y)\}$$

is a map with the two properties. Uniqueness again by induction. ■

5.4 Addition

Definition 5.4.1 (Addition)**.** Let $m \in \mathbb{N}$. By Theorem 5.3.1 (with $A = \mathbb{N}$, $a = m$ and $g = S$) there is exactly one map $h\colon \mathbb{N} \to \mathbb{N}$ with $0 \overset{h}{\longmapsto} m$ and $h \circ S = S \circ h$. *Addition* is the operation on $\mathbb{N}$ (Definition 4.6.1) given by

$$(m, n) \overset{+}{\longmapsto} k \;:\Leftrightarrow\; n \overset{h}{\longmapsto} k \quad \text{for this } h,$$

which exists by Theorem 4.1.3. We write $m + n$ for its value. It is characterised by

$$\text{(A1)} \quad m + 0 = m, \qquad\qquad \text{(A2)} \quad m + (n \cup \{n\}) = (m + n) \cup \{m + n\}.$$

Since $1 = 0 \cup \{0\}$, (A1) and (A2) give $n + 1 = (n + 0) \cup \{n + 0\} = n \cup \{n\}$. So the successor of n is $n + 1$, and by Rule 2 *from now on only* $n + 1$ *is written*. In particular $n \overset{S}{\longmapsto} n + 1$, and (A2) reads $m + (n + 1) = (m + n) + 1$. For example, since $2 = 1 + 1$,

$$2 + 2 = 2 + (1 + 1) = (2 + 1) + 1 = 3 + 1 = 4.$$

Theorem 5.4.2 (Laws of addition)**.** *For all $k, m, n \in \mathbb{N}$:*

(1) $0 + n = n$*;*

(2) $(m + 1) + n = (m + n) + 1$*;*

(3) $m + n = n + m$ *(commutativity);*

(4) $(k + m) + n = k + (m + n)$ *(associativity);*

(5) $k + n = m + n \Rightarrow k = m$ *(cancellation);*

(6) $m + n = 0 \Rightarrow m = 0 \wedge n = 0$*.*

Proof. (1)–(5) by induction on n, using (A1) and (A2) at each step.

(1) $0 + 0 = 0$. If $0 + n = n$, then $0 + (n + 1) = (0 + n) + 1 = n + 1$.

(2) $(m + 1) + 0 = m + 1 = (m + 0) + 1$. If it holds for n, then $(m + 1) + (n + 1) = ((m + 1) + n) + 1 = ((m + n) + 1) + 1 = (m + (n + 1)) + 1$.

(3) $m + 0 = m = 0 + m$ by (1). If $m + n = n + m$, then $m + (n + 1) = (m + n) + 1 = (n + m) + 1 = (n + 1) + m$ by (2).

(4) $(k + m) + 0 = k + m = k + (m + 0)$. If it holds for n, then $(k + m) + (n + 1) = ((k + m) + n) + 1 = (k + (m + n)) + 1 = k + ((m + n) + 1) = k + (m + (n + 1))$.

(5) For $n = 0$ use (A1). Step: $k + (n + 1) = m + (n + 1)$ means $(k + n) + 1 = (m + n) + 1$, so $k + n = m + n$ by Theorem 5.2.4(4), so $k = m$ by the induction hypothesis.

(6) If $n \neq 0$, then $n = p + 1$ (Corollary 5.2.5), and $m + n = (m + p) + 1 \neq 0$ (Theorem 5.2.4(3)). So $n = 0$, and then $m = m + 0 = 0$. ∎

By associativity, brackets in sums like $k + m + n$ can be left out.

5.5 Multiplication

Definition 5.5.1 (Multiplication)**.** Let $m \in \mathbb{N}$, and let $g\colon \mathbb{N} \to \mathbb{N}$ be the map with $x \overset{g}{\longmapsto} x + m$. By Theorem 5.3.1 there is exactly one $h\colon \mathbb{N} \to \mathbb{N}$ with $0 \overset{h}{\longmapsto} 0$ and $h \circ S = g \circ h$. *Multiplication* is the operation on $\mathbb{N}$ with $(m, n) \overset{\cdot}{\longmapsto} k :\Leftrightarrow n \overset{h}{\longmapsto} k$. Its value is written $m \cdot n$. It is characterised by

$$\text{(M1)} \quad m \cdot 0 = 0, \qquad\qquad \text{(M2)} \quad m \cdot (n + 1) = m \cdot n + m.$$

Products are evaluated before sums: $m \cdot n + m$ means $(m \cdot n) + m$.

Theorem 5.5.2 (Laws of multiplication)**.** *For all $k, m, n \in \mathbb{N}$:*

(1) $0 \cdot n = 0$*;*

(2) $(m + 1) \cdot n = m \cdot n + n$*;*

(3) $m \cdot n = n \cdot m$ *(commutativity);*

(4) $k \cdot (m + n) = k \cdot m + k \cdot n$ *and* $(m + n) \cdot k = m \cdot k + n \cdot k$ *(distributivity);*

(5) $(k \cdot m) \cdot n = k \cdot (m \cdot n)$ *(associativity);*

(6) $1 \cdot n = n = n \cdot 1$ *(1 is neutral);*

(7) $m \cdot n = 0 \Rightarrow m = 0 \vee n = 0$*.*

Proof. (1)–(5) by induction on n, using (M1), (M2) and Theorem 5.4.2.
(1) $0 \cdot 0 = 0$. If $0 \cdot n = 0$, then $0 \cdot (n + 1) = 0 \cdot n + 0 = 0$.
(2) $(m + 1) \cdot 0 = 0 = m \cdot 0 + 0$. If it holds for n:

$$\begin{aligned}(m + 1) \cdot (n + 1) &= (m + 1) \cdot n + (m + 1) = (m \cdot n + n) + (m + 1)\\ &= m \cdot n + ((n + m) + 1) = m \cdot n + (m + (n + 1))\\ &= (m \cdot n + m) + (n + 1) = m \cdot (n + 1) + (n + 1).\end{aligned}$$

(3) $m \cdot 0 = 0 = 0 \cdot m$ by (1). If $m \cdot n = n \cdot m$, then $m \cdot (n+1) = m \cdot n + m = n \cdot m + m = (n+1) \cdot m$ by (2).
(4) $k \cdot (m + 0) = k \cdot m = k \cdot m + k \cdot 0$. If it holds for n, then

$$k \cdot (m + (n + 1)) = k \cdot ((m + n) + 1) = k \cdot (m + n) + k = (k \cdot m + k \cdot n) + k = k \cdot m + k \cdot (n + 1).$$

The second form follows by (3).
(5) $(k \cdot m) \cdot 0 = 0 = k \cdot (m \cdot 0)$. If it holds for n, then $(k \cdot m) \cdot (n + 1) = (k \cdot m) \cdot n + k \cdot m = k \cdot (m \cdot n) + k \cdot m = k \cdot (m \cdot n + m) = k \cdot (m \cdot (n + 1))$, using (4).
(6) $n \cdot 1 = n \cdot (0 + 1) = n \cdot 0 + n = n$, and $1 \cdot n = n$ by (3).
(7) If $m \neq 0$ and $n \neq 0$, write $m = p + 1$ and $n = q + 1$ (Corollary 5.2.5). Then $m \cdot n = m \cdot q + (p + 1) = (m \cdot q + p) + 1 \neq 0$. ∎

5.6 Order

Definition 5.6.1 (Order on $\mathbb{N}$)**.** For $m, n \in \mathbb{N}$:

$$m < n :\Leftrightarrow m \in n, \qquad m \leq n :\Leftrightarrow m < n \vee m = n.$$

So $n = \{m \in \mathbb{N} \mid m < n\}$: a natural number is the set of the numbers before it, as promised. (Elements of n are natural numbers by Lemma 5.2.2(1).)

Lemma 5.6.2. *For all $m, n \in \mathbb{N}$:*

(1) $m < n + 1 \Leftrightarrow m \leq n$*;*

(2) $\neg(n < 0)$*;*

(3) $0 \leq n$*;*

(4) $m < n \Rightarrow m + 1 \leq n$*.*

Proof. (1) $n+1 = n \cup \{n\}$, so $m \in n+1 \Leftrightarrow m \in n \vee m = n$.

(2) $0 = \varnothing$ has no elements.

(3) By induction: $0 \leq 0$, and $0 \leq n$ gives $0 < n+1$ by (1).

(4) By induction on n. For $n = 0$ there is nothing to show, by (2). Assume it holds for n and let $m < n+1$, so $m \leq n$ by (1). If $m = n$, then $m+1 = n+1$. If $m < n$, then $m+1 \leq n$ by the induction hypothesis, so $m+1 < n+1$ by (1). ■

Theorem 5.6.3 ($\leq$ is a total order)**.** *$<$ is irreflexive and transitive on $\mathbb{N}$, and* trichotomy *holds: for all m, n exactly one of*

$$m < n, \qquad m = n, \qquad n < m$$

is true. Consequently $\leq$ is a total order on $\mathbb{N}$ (Definition 3.5.1).

Proof. Irreflexive: $n \notin n$ (Lemma 5.2.2(3)). Transitive: if $k \in m$ and $m \in n$, then $m \subseteq n$ (Lemma 5.2.2(2)), so $k \in n$.

At least one holds, by induction on n for fixed m. For $n = 0$: $0 \leq m$ by Lemma 5.6.2(3), so $0 = m$ or $0 < m$. Step: assume one of the three holds for n. If $m < n$ or $m = n$, then $m \leq n$, so $m < n+1$ by Lemma 5.6.2(1). If $n < m$, then $n+1 \leq m$ by Lemma 5.6.2(4). Either way one of the three holds for $n+1$.

At most one holds: $m < n$ together with $m = n$ or with $n < m$ would give $m < m$ by transitivity, against irreflexivity.

By Lemma 3.5.2, $\leq$ is a partial order, and trichotomy makes it total. ■

Theorem 5.6.4 (Order and arithmetic)**.** *For all $k, m, n \in \mathbb{N}$:*

(1) $m \leq n \Leftrightarrow \exists d \in \mathbb{N}\ n = m + d$, and this d is unique;

(2) $m < n \Leftrightarrow \exists d \in \mathbb{N}\ n = m + (d+1)$;

(3) $m < n \Rightarrow m + k < n + k$;

(4) $m < n \wedge k \neq 0 \Rightarrow m \cdot k < n \cdot k$.

Proof. (1) ($\Leftarrow$) By induction on d: $m \leq m + 0$. If $m \leq m + d$, then $m + d < (m + d) + 1 = m + (d+1)$ (Lemma 5.6.2(1)), so $m < m + (d+1)$ by transitivity. ($\Rightarrow$) By induction on n. If $m \leq 0$, then $m = 0$ (Lemma 5.6.2(2)), and $d = 0$ works. Assume the claim for n, and let $m \leq n+1$. If $m = n+1$, take $d = 0$. Otherwise $m < n+1$, so $m \leq n$, so $n = m + d$ for some d, and $n + 1 = (m + d) + 1 = m + (d+1)$. Uniqueness: $m + d = m + d'$ gives $d = d'$ by Theorem 5.4.2(3) and (5).

(2) ($\Rightarrow$) $m + 1 \leq n$ (Lemma 5.6.2(4)), so by (1) $n = (m+1) + d = m + (d+1)$, using Theorem 5.4.2. ($\Leftarrow$) $m \leq m + d < (m + d) + 1 = n$ by (1) and Lemma 5.6.2(1).

(3) By (2), $n = m + (d+1)$, so $n + k = (m + k) + (d+1)$ by Theorem 5.4.2(3) and (4). Use (2) again.

(4) By (2), $n = m + (d+1)$, so $n \cdot k = m \cdot k + (d+1) \cdot k$. Since $k \neq 0$ and $d + 1 \neq 0$, $(d+1) \cdot k \neq 0$ (Theorem 5.5.2(7)), so $(d+1) \cdot k = e + 1$ for some e (Corollary 5.2.5). Use (2). ■

Corollary 5.6.5 (Cancellation)**.** *$m+k < n+k \Rightarrow m < n$. And if $k \neq 0$: $m \cdot k = n \cdot k \Rightarrow m = n$, and $m \cdot k < n \cdot k \Rightarrow m < n$.*

Proof. Each time, compare m and n by trichotomy (Theorem 5.6.3). The cases other than the claimed one contradict the hypothesis by Theorem 5.6.4(3) and (4). For example, if $m \cdot k = n \cdot k$ but $m < n$, then $m \cdot k < n \cdot k$, contradicting irreflexivity. ■

5.7 Well-ordering

Theorem 5.7.1 (Well-ordering principle)**.** *Every nonempty subset of* $\mathbb{N}$ *has a least element. So* $(\mathbb{N}, \leq)$ *is well-ordered (Definition 3.5.5).*

Proof. Let $X \subseteq \mathbb{N}$ have no least element. We show $X = \varnothing$. Let

$$Y \coloneqq \{n \in \mathbb{N} \mid \forall m \in \mathbb{N}\ (m < n \Rightarrow m \notin X)\}$$

be the set of n such that no number below n is in X. We show Y is inductive. $0 \in Y$ vacuously (Lemma 5.6.2(2)). Let $n \in Y$. If n were in X, it would be the least element of X: for $x \in X$, $x < n$ is impossible because $n \in Y$, so $n \leq x$ by trichotomy. So $n \notin X$. Then no $m \leq n$ is in X, so no $m < n + 1$ (Lemma 5.6.2(1)). So $n + 1 \in Y$.

By induction $Y = \mathbb{N}$. Every $n \in \mathbb{N}$ satisfies $n < n + 1$ and $n + 1 \in Y$, so $n \notin X$. ∎

Corollary 5.7.2 (Strong induction)**.** *Let* P *be a predicate such that for every* $n \in \mathbb{N}$*,*

$$\big(\forall m \in \mathbb{N}\ (m < n \Rightarrow P(m))\big) \Rightarrow P(n).$$

Then $P(n)$ *holds for every* $n \in \mathbb{N}$*.*

Proof. Suppose $X \coloneqq \{n \in \mathbb{N} \mid \neg P(n)\}$ is nonempty, and let n be its least element (Theorem 5.7.1). Every $m < n$ is outside X, so $P(m)$ holds for all $m < n$. The hypothesis gives $P(n)$, contradicting $n \in X$. ∎

The case $n = 0$ has no $m < 0$, so it requires proving $P(0)$ outright. Strong induction lets the step use *all* smaller cases, not only the previous one. It will be used for prime factorization.

5.8 What is still missing

$\mathbb{N}$ has addition and multiplication with all the familiar laws, and an order compatible with both. But the equation $m + x = n$ only has a solution $x \in \mathbb{N}$ when $m \leq n$ (Theorem 5.6.4(1)). There is no x with $1 + x = 0$. Chapter 6 builds a larger number system where subtraction always works.

Exercises

1. Prove $2 \cdot 3 = 6$ from the definitions, citing each step.
2. Prove that there is no m with $n < m$ and $m < n + 1$.
3. Use Theorem 5.3.1 to define, for each m, the map $p\colon \mathbb{N} \to \mathbb{N}$ with $0 \overset{p}{\longmapsto} 1$ and $n \overset{p}{\longmapsto} y \Rightarrow n + 1 \overset{p}{\longmapsto} y \cdot m$. Its value at n is the n-th power of m. Show that the value at $a + b$ is the product of the values at a and at b.
4. Use Corollary 5.3.2 to define the *factorial*, the map $F\colon \mathbb{N} \to \mathbb{N}$ with $0 \overset{F}{\longmapsto} 1$ and $n \overset{F}{\longmapsto} y \Rightarrow n + 1 \overset{F}{\longmapsto} y \cdot (n + 1)$. Find the value of F at 4.
5. For a sequence $x\colon \mathbb{N} \to \mathbb{N}$, use Corollary 5.3.2 to define its *partial sums*, the sequence s with $0 \overset{s}{\longmapsto} 0$ and $n \overset{s}{\longmapsto} y \wedge n \overset{x}{\longmapsto} z \Rightarrow n + 1 \overset{s}{\longmapsto} y + z$. Let $x = \mathrm{id}_{\mathbb{N}}$. Prove: if $n + 1 \overset{s}{\longmapsto} y$, then $2 \cdot y = n \cdot (n + 1)$.
6. Show that $\mathbb{N}$ has no greatest element.

Chapter 6

Integers

In $\mathbb{N}$, $3 - 5$ doesn't exist. We build a larger system where it does.

The idea: every integer should be a difference $a - b$ of natural numbers, but we can't write $a - b$ yet. So we use the pair (a, b) as a stand-in for "$a - b$". Different pairs stand for the same difference: $(0, 2)$, $(1, 3)$ and $(5, 7)$ all mean "-2". When do (a, b) and (c, d) mean the same thing? When $a - b = c - d$, that is, when $a + d = c + b$. The last equation uses only addition, which we have. So we *define* an integer as a class of pairs under that equation.

6.1 Construction

Definition 6.1.1. On $\mathbb{N} \times \mathbb{N}$ let

$$(a, b) \sim (c, d) :\Leftrightarrow a + d = c + b.$$

Lemma 6.1.2. *$\sim$ is an equivalence relation.*

Proof. Reflexive: $a + b = a + b$. Symmetric: $a + d = c + b \Rightarrow c + b = a + d$. Transitive: let $a + d = c + b$ and $c + f = e + d$. Adding the equations and rearranging (Theorem 5.4.2) gives $(a + f) + (c + d) = (e + b) + (c + d)$. Cancelling $c + d$ (Theorem 5.4.2(5)) gives $a + f = e + b$, so $(a, b) \sim (e, f)$. ∎

Definition 6.1.3 (Integers)**.** $\mathbb{Z} \coloneqq (\mathbb{N} \times \mathbb{N})/\sim$. Its elements are the *integers*. The class of (a, b) is $[(a, b)]$; think of it as "$a - b$".

By Lemma 3.4.2, $[(a, b)] = [(c, d)] \Leftrightarrow a + d = c + b$.

6.2 Arithmetic

The definitions below come from asking what $(a - b) + (c - d)$, $-(a - b)$ and $(a - b) \cdot (c - d)$ *would* be, and writing the answer as a difference without using subtraction.

Definition 6.2.1 (Operations on $\mathbb{Z}$)**.**

$$\begin{aligned} [(a, b)] + [(c, d)] &\coloneqq [(a + c,\ b + d)], \\ -[(a, b)] &\coloneqq [(b, a)], \\ [(a, b)] \cdot [(c, d)] &\coloneqq [(a \cdot c + b \cdot d,\ a \cdot d + b \cdot c)], \end{aligned}$$

with $0 \coloneqq [(0, 0)]$ and $1 \coloneqq [(1, 0)]$. *Subtraction* is $x - y \coloneqq x + (-y)$. The symbols $+$, $\cdot$, 0 and 1 are the same as in $\mathbb{N}$; Theorem 6.3.1 shows this is consistent (Rule 8).

These use representatives, so they are only valid definitions once they're shown to be well-defined (Definition 3.4.5 and Theorem 4.1.3).

Lemma 6.2.2. *The right-hand sides in Definition 6.2.1 don't depend on the choice of representatives.*

Proof. Let $(a,b) \sim (a',b')$, so $a+b' = a'+b$, and $(c,d) \sim (c',d')$, so $c+d' = c'+d$.

Addition. We need $(a+c)+(b'+d') = (a'+c')+(b+d)$. That is the sum of the two hypotheses, rearranged.

Negation. We need $b+a' = b'+a$. That is the first hypothesis, read backwards.

Multiplication. First change only the first factor. We need

$$(a\cdot c+b\cdot d)+(a'\cdot d+b'\cdot c) = (a'\cdot c+b'\cdot d)+(a\cdot d+b\cdot c).$$

By Theorems 5.4.2 and 5.5.2, the left side is $(a+b')\cdot c+(b+a')\cdot d$ and the right side is $(a'+b)\cdot c+(b'+a)\cdot d$. These are equal because $a+b' = a'+b$. The formula is symmetric in the two pairs, so the same argument covers changing the second factor. Changing both is one change after the other. ■

Definition 6.2.3 (Commutative ring)**.** A *commutative ring* is a set R with two binary operations $+$ and $\cdot$, a unary operation $-$, and two elements $0,1 \in R$ such that for all $x,y,z \in R$:

	Addition	Multiplication
Associative	$(x+y)+z = x+(y+z)$	$(x\cdot y)\cdot z = x\cdot(y\cdot z)$
Commutative	$x+y = y+x$	$x\cdot y = y\cdot x$
Neutral element	$x+0 = x$	$x\cdot 1 = x$
Inverse	$x+(-x) = 0$	(not required)
Distributive	$x\cdot(y+z) = x\cdot y + x\cdot z$	

Subtraction is $x-y \coloneqq x+(-y)$.

$\mathbb{N}$ is not a commutative ring, since additive inverses are missing.

Theorem 6.2.4. *$\mathbb{Z}$, with the operations of Definition 6.2.1, is a commutative ring, and $0 \neq 1$.*

Proof. Write $x = [(a,b)]$, $y = [(c,d)]$ and $z = [(e,f)]$. Every law reduces to laws of $\mathbb{N}$ (Theorems 5.4.2 and 5.5.2) applied to the components.

Addition is associative and commutative because it is componentwise. $x+0 = [(a+0, b+0)] = x$. $x+(-x) = [(a+b, b+a)] = [(0,0)]$, since $(a+b)+0 = 0+(b+a)$.

Multiplication. Commutative: the formula is symmetric in the two pairs. Neutral: $x\cdot 1 = [(a\cdot 1+b\cdot 0,\ a\cdot 0+b\cdot 1)] = [(a,b)]$. Associative: expanding with distributivity in $\mathbb{N}$, $(x\cdot y)\cdot z$ and $x\cdot(y\cdot z)$ both equal

$$[(a\cdot c\cdot e+a\cdot d\cdot f+b\cdot c\cdot f+b\cdot d\cdot e,\ a\cdot c\cdot f+a\cdot d\cdot e+b\cdot c\cdot e+b\cdot d\cdot f)].$$

Distributivity. Both $x\cdot(y+z)$ and $x\cdot y+x\cdot z$ equal

$$[(a\cdot c+a\cdot e+b\cdot d+b\cdot f,\ a\cdot d+a\cdot f+b\cdot c+b\cdot e)].$$

$0 \neq 1$: $[(0,0)] = [(1,0)]$ would mean $0+0 = 1+0$, so $0 = 0+1$, against Theorem 5.2.4(3). ■

Lemma 6.2.5 (Consequences of the ring laws)**.** *In every commutative ring:*

(1) the additive inverse is unique: if $x + y = 0$, then $y = -x$;

(2) $x \cdot 0 = 0$;

(3) $(-x) \cdot y = -(x \cdot y)$ and $(-x) \cdot (-y) = x \cdot y$;

(4) $-(-x) = x$ and $-(x + y) = (-x) + (-y)$;

(5) $x + z = y + z \Rightarrow x = y$.

Proof. (1) $y = y + (x + (-x)) = (y + x) + (-x) = 0 + (-x) = -x$.

(2) $x \cdot 0 = x \cdot (0 + 0) = x \cdot 0 + x \cdot 0$. Adding $-(x \cdot 0)$ to both sides gives $0 = x \cdot 0$.

(4) $(-x) + x = 0$, so $x = -(-x)$ by (1). And $(x + y) + ((-x) + (-y)) = 0$, so (1) applies again.

(3) $(-x) \cdot y + x \cdot y = ((-x) + x) \cdot y = 0 \cdot y = 0$, so $(-x) \cdot y = -(x \cdot y)$ by (1). Then $(-x) \cdot (-y) = -(x \cdot (-y)) = -(-(x \cdot y)) = x \cdot y$ by (4).

(5) Add $-z$ to both sides. ■

Since only the ring laws are used, Lemma 6.2.5 holds equally for $\mathbb{Q}$, and later for $\mathbb{R}$, the complex numbers and rings of matrices. Proving things *once, from the laws*, is the main gain of naming a structure.

6.3 $\mathbb{N}$ inside $\mathbb{Z}$

Theorem 6.3.1 (Embedding)**.** *Let $\iota\colon \mathbb{N} \to \mathbb{Z}$ be the map with $n \overset{\iota}{\longmapsto} [(n, 0)]$. Then ι is injective, $0 \overset{\iota}{\longmapsto} 0$, $1 \overset{\iota}{\longmapsto} 1$, and for all $m, n \in \mathbb{N}$ and $x, y \in \mathbb{Z}$:*

$$m \overset{\iota}{\longmapsto} x \wedge n \overset{\iota}{\longmapsto} y \;\Rightarrow\; m + n \overset{\iota}{\longmapsto} x + y \;\wedge\; m \cdot n \overset{\iota}{\longmapsto} x \cdot y \;\wedge\; [(m, n)] = x - y.$$

Proof. Injective: $[(m, 0)] = [(n, 0)]$ means $m + 0 = n + 0$. Next, $[(m, 0)] + [(n, 0)] = [(m + n, 0)]$ and $[(m, 0)] \cdot [(n, 0)] = [(m \cdot n + 0,\ m \cdot 0 + 0 \cdot n)] = [(m \cdot n, 0)]$. Finally $[(m, 0)] - [(n, 0)] = [(m, 0)] + [(0, n)] = [(m, n)]$. ■

So ι copies $\mathbb{N}$, with its arithmetic, into $\mathbb{Z}$. By Rule 8 *we identify every $n \in \mathbb{N}$ with the integer* $[(n, 0)]$ and treat $\mathbb{N}$ as a subset of $\mathbb{Z}$. Strictly speaking, the set $2 = \{0, 1\}$ and the class $[(2, 0)]$ are different sets, but nothing we do depends on that, because ι preserves all the structure. Under the identification the construction says what we wanted: $[(a, b)] = a - b$. By Rule 2, from now on integers are written with $-$ and the class notation is not used again.

Corollary 6.3.2 (Normal form)**.** *Every integer x satisfies exactly one of*

$$x = n \text{ for some } n \in \mathbb{N} \setminus \{0\}, \qquad x = 0, \qquad x = -n \text{ for some } n \in \mathbb{N} \setminus \{0\},$$

and n is unique.

Proof. Write $x = a - b$ with $a, b \in \mathbb{N}$. If $b \leq a$, then $a = b + d$ for some $d \in \mathbb{N}$ (Theorem 5.6.4(1)), and $x = (b + d) - b = d$. If $a < b$, then $b = a + d$ with $d \neq 0$, and $x = a - (a + d) = -d$. The cases are exclusive and n is unique: $n = m$ in $\mathbb{Z}$ gives $n = m$ in $\mathbb{N}$ because ι is injective, and $n = -m$ gives $n + m = 0$, so $n = m = 0$ (Theorem 5.4.2(6)). ■

6.4 No zero divisors

Theorem 6.4.1. *For $x, y, z \in \mathbb{Z}$:*

(1) $x \cdot y = 0 \Rightarrow x = 0 \vee y = 0$*;*

(2) $x \cdot z = y \cdot z \wedge z \neq 0 \Rightarrow x = y$ *(cancellation).*

A commutative ring with $0 \neq 1$ *and property (1) is called an* integral domain*.*

Proof. (1) By Corollary 6.3.2, x is m or $-m$ and y is n or $-n$ for some $m, n \in \mathbb{N}$. By Lemma 6.2.5(3), $x \cdot y$ is $m \cdot n$ or $-(m \cdot n)$. If $x \cdot y = 0$, then $m \cdot n = 0$, because $-k = 0$ only for $k = 0$. So $m = 0$ or $n = 0$ (Theorem 5.5.2(7)).

(2) $(x - y) \cdot z = x \cdot z - y \cdot z = 0$ by distributivity and Lemma 6.2.5(3). Since $z \neq 0$, (1) gives $x - y = 0$, so $x = y$. ■

6.5 Order

Definition 6.5.1 (Order on $\mathbb{Z}$)**.** $x \leq y :\Leftrightarrow y - x \in \mathbb{N}$. An integer x is *positive* if $0 < x$ and *negative* if $x < 0$.

Theorem 6.5.2 ($\mathbb{Z}$ is an ordered ring)**.**

(1) $\leq$ *is a total order on* $\mathbb{Z}$*;*

(2) $x \leq y \Rightarrow x + z \leq y + z$*;*

(3) $0 \leq x \wedge 0 \leq y \Rightarrow 0 \leq x \cdot y$*;*

(4) on $\mathbb{N}$*, this order agrees with the order of Chapter 5.*

Proof. We use three facts: $\mathbb{N}$ is closed under $+$ and $\cdot$ (Theorem 6.3.1); for every x, $x \in \mathbb{N}$ or $-x \in \mathbb{N}$; and $n \in \mathbb{N}$ together with $-n \in \mathbb{N}$ forces $n = 0$ (both by Corollary 6.3.2).

(1) Reflexive: $x - x = 0 \in \mathbb{N}$. Antisymmetric: if $y - x \in \mathbb{N}$ and $x - y = -(y - x) \in \mathbb{N}$, then $y - x = 0$. Transitive: $z - x = (z - y) + (y - x)$. Total: $y - x$ or $-(y - x) = x - y$ lies in $\mathbb{N}$.

(2) $(y + z) - (x + z) = y - x$.

(3) $x \cdot y \in \mathbb{N}$ when $x, y \in \mathbb{N}$.

(4) For $m, n \in \mathbb{N}$: $m \leq n$ in $\mathbb{N}$ if and only if $n = m + d$ for some $d \in \mathbb{N}$ (Theorem 5.6.4(1)), if and only if $n - m \in \mathbb{N}$. ■

Familiar rules follow from Theorem 6.5.2: for example $x \leq y \Leftrightarrow -y \leq -x$, and $x \leq y \wedge 0 \leq z \Rightarrow x \cdot z \leq y \cdot z$ (Exercise 3).

6.6 What is still missing

In $\mathbb{Z}$ every equation $a + x = b$ has exactly one solution, $x = b - a$. But $2 \cdot x = 1$ has none: $2 \cdot x$ is 0 for $x = 0$, at least 2 for $0 < x$, and negative for $x < 0$. Chapter 7 adds the missing quotients.

Exercises

1. Show that for $a, b, c, d \in \mathbb{N}$: $a - b \leq c - d \Leftrightarrow a + d \leq c + b$.

2. Let $|\ |$ be the unary operation on $\mathbb{Z}$ with $|x| = x$ if $0 \leq x$ and $|x| = -x$ otherwise. Prove $|x \cdot y| = |x| \cdot |y|$ and $|x + y| \leq |x| + |y|$ (the *triangle inequality*).

3. Prove from Theorem 6.5.2: $x \leq y \Rightarrow -y \leq -x$; $x \leq y \wedge 0 \leq z \Rightarrow x \cdot z \leq y \cdot z$; $0 \leq x \cdot x$ for all x; $0 < 1$.

4. Why does the construction need *ordered* pairs? Try using unordered pairs $\{a, b\}$ instead of (a, b) and find where it breaks. (Compare "$3 - 5$" and "$5 - 3$".)

Chapter 7

Rational numbers

In $\mathbb{Z}$, $2 \cdot x = 1$ has no solution. We repeat the trick of Chapter 6, now for division: a rational number is what a quotient of integers *would* be. As a stand-in we use the pair (a, b) with $b \neq 0$. Two pairs mean the same quotient when "a over b" equals "c over d", that is, when $a \cdot d = c \cdot b$. That equation only uses multiplication, which we have.

7.1 Construction

Definition 7.1.1. Let $\mathbb{Z}^* \coloneqq \mathbb{Z} \setminus \{0\}$. On $\mathbb{Z} \times \mathbb{Z}^*$ let

$$(a, b) \approx (c, d) :\Leftrightarrow a \cdot d = c \cdot b.$$

Lemma 7.1.2. *$\approx$ is an equivalence relation.*

Proof. Reflexive and symmetric are clear. Transitive: let $a \cdot d = c \cdot b$ and $c \cdot f = e \cdot d$. Multiplying the first by f gives $a \cdot d \cdot f = c \cdot b \cdot f = (c \cdot f) \cdot b = e \cdot d \cdot b$, so $(a \cdot f) \cdot d = (e \cdot b) \cdot d$. Since $d \neq 0$, cancellation in $\mathbb{Z}$ (Theorem 6.4.1(2)) gives $a \cdot f = e \cdot b$. ■

This is where Chapter 6 pays off: without "no zero divisors", $\approx$ would not be transitive.

Definition 7.1.3 (Rational numbers)**.** $\mathbb{Q} \coloneqq (\mathbb{Z} \times \mathbb{Z}^*)/\!\approx$. Its elements are the *rational numbers*. The class of (a, b) is $[(a, b)]$, with *numerator* a and *denominator* b.

So $[(a, b)] = [(c, d)] \Leftrightarrow a \cdot d = c \cdot b$.

Lemma 7.1.4 (Expanding)**.** $[(a \cdot k, b \cdot k)] = [(a, b)]$ *for* $k \neq 0$. *In particular* $[(-a, -b)] = [(a, b)]$, *so every rational number has a representative whose denominator lies in* $\mathbb{N}$.

Proof. $(a \cdot k) \cdot b = a \cdot (b \cdot k)$. Take $k = -1$ for the second claim, and use Corollary 6.3.2 for the third. ■

7.2 Arithmetic

Definition 7.2.1 (Operations on $\mathbb{Q}$)**.**

$$\begin{aligned} [(a, b)] + [(c, d)] &\coloneqq [(a \cdot d + c \cdot b,\ b \cdot d)], \\ -[(a, b)] &\coloneqq [(-a, b)], \\ [(a, b)] \cdot [(c, d)] &\coloneqq [(a \cdot c,\ b \cdot d)], \end{aligned}$$

with $0 \coloneqq [(0, 1)]$ and $1 \coloneqq [(1, 1)]$. The denominators are allowed: $b, d \neq 0$ gives $b \cdot d \neq 0$ by Theorem 6.4.1(1). As in $\mathbb{Z}$, the symbols are shared with the smaller number systems (Rule 8 and Theorem 7.3.1).

Lemma 7.2.2. *The operations of Definition 7.2.1 are well-defined.*

Proof. Let $a \cdot b' = a' \cdot b$. We change the first argument; both formulas are symmetric in the two arguments, so that is enough.

Addition. We need $(a \cdot d + c \cdot b) \cdot (b' \cdot d) = (a' \cdot d + c \cdot b') \cdot (b \cdot d)$. Expanded, the left side is $a \cdot b' \cdot d \cdot d + c \cdot b \cdot b' \cdot d$ and the right side is $a' \cdot b \cdot d \cdot d + c \cdot b' \cdot b \cdot d$. They are equal because $a \cdot b' = a' \cdot b$.

Multiplication. $(a \cdot c) \cdot (b' \cdot d) = (a \cdot b') \cdot c \cdot d = (a' \cdot b) \cdot c \cdot d = (a' \cdot c) \cdot (b \cdot d)$.

Negation. $(-a) \cdot b' = -(a \cdot b') = -(a' \cdot b) = (-a') \cdot b$. ■

Definition 7.2.3 (Field)**.** A *field* is a commutative ring with $0 \neq 1$ in which every $x \neq 0$ has a *multiplicative inverse* x^{-1} with $x \cdot x^{-1} = 1$. For $y \neq 0$, the *quotient* is $x/y \coloneqq x \cdot y^{-1}$.

Theorem 7.2.4. *$\mathbb{Q}$ is a field. The inverse of $[(a,b)] \neq 0$ is $[(b,a)]$.*

Proof. Let $x = [(a,b)]$, $y = [(c,d)]$ and $z = [(e,f)]$. Commutativity of both operations is built into the formulas.

Associativity. $(x+y)+z$ and $x+(y+z)$ both equal $[(a \cdot d \cdot f + c \cdot b \cdot f + e \cdot b \cdot d,\ b \cdot d \cdot f)]$, and $(x \cdot y) \cdot z$ and $x \cdot (y \cdot z)$ both equal $[(a \cdot c \cdot e,\ b \cdot d \cdot f)]$.

Neutral elements. $x + 0 = [(a \cdot 1 + 0 \cdot b,\ b \cdot 1)] = x$, and $x \cdot 1 = [(a \cdot 1,\ b \cdot 1)] = x$.

Additive inverse. $x + (-x) = [(a \cdot b + (-a) \cdot b,\ b \cdot b)] = [(0,\ b \cdot b)] = [(0,1)]$, since $0 \cdot 1 = 0 \cdot (b \cdot b)$.

Distributivity. Using Lemma 7.1.4 with $k = b$:

$$x \cdot y + x \cdot z = [(a \cdot c \cdot b \cdot f + a \cdot e \cdot b \cdot d,\ b \cdot d \cdot b \cdot f)] = [(a \cdot (c \cdot f + e \cdot d),\ b \cdot d \cdot f)] = x \cdot (y+z).$$

$0 \neq 1$. $[(0,1)] = [(1,1)]$ would mean $0 = 1$ in $\mathbb{Z}$.

Multiplicative inverse. First, $[(a,b)] = 0$ if and only if $a \cdot 1 = 0 \cdot b$, if and only if $a = 0$. So if $x \neq 0$, then $a \neq 0$, and $[(b,a)]$ is a rational number with $[(a,b)] \cdot [(b,a)] = [(a \cdot b,\ b \cdot a)] = [(1,1)]$. ■

All of Lemma 6.2.5 holds in $\mathbb{Q}$, since only ring laws were used there. A field has no zero divisors: if $x \cdot y = 0$ and $x \neq 0$, then $y = x^{-1} \cdot x \cdot y = 0$.

7.3 $\mathbb{Z}$ inside $\mathbb{Q}$

Theorem 7.3.1 (Embedding)**.** *Let $j \colon \mathbb{Z} \to \mathbb{Q}$ be the map with $a \overset{j}{\longmapsto} [(a,1)]$. Then j is injective, $0 \overset{j}{\longmapsto} 0$, $1 \overset{j}{\longmapsto} 1$, and for all $a, c \in \mathbb{Z}$, $b \in \mathbb{Z}^*$ and $x, y, u \in \mathbb{Q}$:*

$$a \overset{j}{\longmapsto} x \wedge c \overset{j}{\longmapsto} y \ \Rightarrow\ a + c \overset{j}{\longmapsto} x + y \ \wedge\ a \cdot c \overset{j}{\longmapsto} x \cdot y, \qquad a \overset{j}{\longmapsto} x \wedge b \overset{j}{\longmapsto} u \ \Rightarrow\ [(a,b)] = x/u.$$

Proof. $[(a,1)] = [(a',1)]$ means $a = a'$. Next, $[(a,1)]+[(c,1)] = [(a+c,1)]$ and $[(a,1)] \cdot [(c,1)] = [(a \cdot c, 1)]$. Finally the inverse of $[(b,1)]$ is $[(1,b)]$ by Theorem 7.2.4, and $[(a,1)] \cdot [(1,b)] = [(a,b)]$. ■

As before, *we identify every $a \in \mathbb{Z}$ with* $[(a,1)]$, so $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$ (Rule 8). The theorem then says $[(a,b)] = a/b$: the class is the quotient. By Rule 2, from now on rationals are written as quotients and the class notation is not used again. In this notation, $a/b = c/d \Leftrightarrow a \cdot d = c \cdot b$.

7.4 Order

Definition 7.4.1 (Order on $\mathbb{Q}$)**.** A rational number is *nonnegative* if it equals a/b for some $a, b \in \mathbb{N}$ with $b \neq 0$. For $x, y \in \mathbb{Q}$ let

$$x \leq y :\Leftrightarrow y - x \text{ is nonnegative.}$$

Theorem 7.4.2 ($\mathbb{Q}$ is an ordered field)**.** *$\leq$ is a total order on $\mathbb{Q}$; $x \leq y \Rightarrow x + z \leq y + z$; and $0 \leq x \wedge 0 \leq y \Rightarrow 0 \leq x \cdot y$. On $\mathbb{Z}$ it agrees with the order of Chapter 6.*

Proof. $0 \leq x$ means that x is nonnegative. The nonnegative rationals have three properties:

- *They are closed under $+$ and $\cdot$:* for $a, b, c, d \in \mathbb{N}$ with $b, d \neq 0$, both $a/b + c/d = (a \cdot d + c \cdot b)/(b \cdot d)$ and $(a/b) \cdot (c/d) = (a \cdot c)/(b \cdot d)$ are quotients of natural numbers.
- *For every x, x or $-x$ is nonnegative:* write $x = a/b$ with $b \in \mathbb{N}$, $b \neq 0$ (Lemma 7.1.4). If $a \in \mathbb{N}$, x is nonnegative; otherwise $-a \in \mathbb{N}$ and $-x = (-a)/b$ is.
- *Only 0 is both:* if $x = a/b$ and $-x = c/d$ with $a, b, c, d \in \mathbb{N}$, then $a/b = (-c)/d$, so $a \cdot d = -(c \cdot b)$, so $a \cdot d + c \cdot b = 0$ in $\mathbb{N}$. Then $a \cdot d = 0$ (Theorem 5.4.2(6)), so $a = 0$ and $x = 0$.

The proof of Theorem 6.5.2(1)–(3) used only these three properties of $\mathbb{N} \subseteq \mathbb{Z}$, so it goes through word for word.

On $\mathbb{Z}$: if $a \in \mathbb{N}$, then $a = a/1$ is nonnegative. Conversely, if $a \in \mathbb{Z}$ and $a = c/d$ with $c, d \in \mathbb{N}$, $d \neq 0$, then $a \cdot d = c \in \mathbb{N}$. If a were negative, $-a \in \mathbb{N} \setminus \{0\}$ and $(-a) \cdot d = -c$ would be a nonzero natural number with $c + (-a) \cdot d = 0$, which is impossible (Theorem 5.4.2(6)). So $a \in \mathbb{N}$. ∎

7.5 Two properties of $\mathbb{Q}$

Theorem 7.5.1 (Archimedean property)**.** *For every $x \in \mathbb{Q}$ there is an $n \in \mathbb{N}$ with $x < n$.*

Proof. Write $x = a/b$ with $b \in \mathbb{N}$, $b \neq 0$, so $1 \leq b$. If $a \leq 0$, then $x \leq 0 < 1$. If $0 < a$, take $n := a + 1$. Then

$$n - x = \big((a+1) \cdot b - a\big)/b = \big(a \cdot (b-1) + b\big)/b,$$

a quotient of natural numbers whose numerator is not 0. So $x < n$. ∎

Theorem 7.5.2 (Density)**.** *If $x < y$ in $\mathbb{Q}$, then $x < (x+y)/2$ and $(x+y)/2 < y$. So between any two rational numbers there is another one, and hence infinitely many.*

Proof. $0 < 2$, so $0 < 2^{-1}$. From $x < y$ we get $2 \cdot x = x + x < x + y$ and $x + y < y + y = 2 \cdot y$. Multiply by 2^{-1}. ∎

7.6 The gap

$\mathbb{Q}$ looks complete: it is a field, it is ordered, and it is dense. But it has holes.

Theorem 7.6.1 (The square root of 2 is irrational)**.** *There is no $q \in \mathbb{Q}$ with $q \cdot q = 2$.*

Proof. Suppose $q \cdot q = 2$. Since $(-q) \cdot (-q) = q \cdot q$, we may assume $0 < q$. Then $1 < q$ and $q < 2$: $q \leq 1$ would give $q \cdot q \leq q \leq 1$, and $2 \leq q$ would give $4 \leq q \cdot q$.

Write $q = a/b$ with $a, b \in \mathbb{N}$, $b \neq 0$. Then $b \cdot q = a$ is a natural number, so the set

$$X \coloneqq \{m \in \mathbb{N} \mid m \neq 0 \wedge m \cdot q \in \mathbb{N}\}$$

is nonempty. By Theorem 5.7.1 it has a least element m. Let $m' \coloneqq m \cdot q - m = m \cdot (q - 1)$. Then:

- m' is an integer, as a difference of two natural numbers, and $0 < m'$ because $1 < q$. So $m' \in \mathbb{N}$ and $m' \neq 0$.
- $m' \cdot q = m \cdot q \cdot q - m \cdot q = 2 \cdot m - m \cdot q$ is an integer, and $0 < m' \cdot q$ because $0 < m'$ and $0 < q$. So $m' \cdot q \in \mathbb{N}$.
- $m' < m$, because $q - 1 < 1$.

So $m' \in X$ and $m' < m$, contradicting the choice of m. ■

The proof uses only order and well-ordering; it needs no divisibility. So the set $\{q \in \mathbb{Q} \mid 0 < q \wedge q \cdot q < 2\}$ is bounded above (by 2), but its least upper bound would have to be a square root of 2, and there is none in $\mathbb{Q}$.

This is where the real numbers come in. They are constructed by filling exactly these gaps, using sequences of rational numbers, i.e. maps $\mathbb{N} \to \mathbb{Q}$, or cuts of $\mathbb{Q}$. From there come limits, series, functions and calculus.

7.7 Summary of the number constructions

Set	Built from	Element	Becomes	Gains
$\mathbb{N}$	$\varnothing$ and successors	$\{0, 1, \ldots\}$	n	counting, induction, recursion
$\mathbb{Z}$	classes of pairs in $\mathbb{N} \times \mathbb{N}$	$[(a, b)]$	$a - b$	subtraction
$\mathbb{Q}$	classes of pairs in $\mathbb{Z} \times \mathbb{Z}^*$	$[(a, b)]$	a/b	division
$\mathbb{R}$	still to come			limits, completeness

Exercises

1. Show that $\mathbb{Q}$ is the smallest field containing $\mathbb{Z}$, in this sense: if F is a field and $\varphi\colon \mathbb{Z} \to F$ is injective and preserves $+$, $\cdot$ and 1, then there is exactly one injective map $\psi\colon \mathbb{Q} \to F$ that preserves $+$ and $\cdot$ and agrees with φ on $\mathbb{Z}$.
2. Show that for $x, y \in \mathbb{Q}$ with $0 < x$ there is $n \in \mathbb{N}$ with $y < n \cdot x$.
3. Adapt the proof of Theorem 7.6.1 to show there is no rational q with $q \cdot q = 3$.
4. Let $S \coloneqq \{q \in \mathbb{Q} \mid 0 < q \wedge q \cdot q < 2\}$. Show that for every $s \in S$ there is a larger element of S. (Try $s + h$ with a suitable small rational $0 < h$.)

Part II

Induction

Introduction to induction

A deductive science constructs a world from axioms and never has to look outside. An inductive science works the other way round: it starts with the world as we find it and asks which constructed world *fits* it. Here we observe, and then construct a model that matches what we observed.

Two meanings of "induction"

The word is used for two unrelated things, and the difference matters here. *Mathematical induction* (Theorem 5.2.1) is a deductive proof method: from $P(0)$ and $P(n) \Rightarrow P(n+1)$ it proves $P(n)$ for *all* n, with certainty. *Scientific induction*, the subject of this part, goes from finitely many observations to a general model. That step is never certain. No number of white swans proves that all swans are white, and a single black one refutes it.

The inductive cycle

Every inductive science goes through the same cycle:

(1) **Observe.** Record what happens, together with how precisely it was measured.

(2) **Model.** Propose a mathematical structure that is meant to describe it. The model is itself a deductive object: a set with relations, maps and numbers, built with the tools of Part I.

(3) **Deduce.** Work out, by proof or computation, what the model predicts, including things nobody has observed yet.

(4) **Compare.** Check the predictions against new observations. Keep the model if it fits, revise or replace it if it doesn't, and go back to step 1.

Deduction is part of every round: step 3 is pure deduction. What makes the science inductive is steps 1 and 4, where the world gets the last word.

What a model can claim

A deductive theorem is true relative to its axioms, once and for all. An inductive model has a weaker and different status:

- It is never proved, only *supported*: it has survived the comparisons made so far.
- It can be *refuted* by a single reliable observation it can't accommodate. A model that no conceivable observation could contradict says nothing about the world.

- It has a *domain of validity*. A model can be excellent at one scale and wrong at another, and still be the right tool inside its domain.
- "Fits" has to be made precise. Observations carry uncertainty, so comparing them with predictions needs *probability* and *statistics*. Probability theory is itself deductive. Statistical inference, which uses it to judge models against data, is the bridge between the two parts of this book.

One science, not many

Physics, chemistry, biology and psychology are usually taught as separate subjects. Here they are not separated, because they follow the same cycle and differ only in the scale and kind of what they observe. A molecule is described by a model built from models of atoms, a cell by models of molecules, behaviour by models of nervous systems. So this part is ordered the way the deductive part is: *nothing is used before it has been introduced*. Here that applies to models and to the observations supporting them, as well as to definitions.

Conventions for this part

Each model will be presented with:

- the **observations** it is meant to explain;
- its **assumptions**, stated as precisely as axioms;
- the **mathematics** it uses, with references to Part I;
- its **predictions**, and how they were tested;
- its known **limits**: where it fails, and which model takes over.

Planned chapters

- measurement, units and uncertainty;
- probability and statistical inference: how to judge a model against data;
- motion and forces;
- matter: from atoms to molecules and reactions;
- life: from molecules to cells, organisms and evolution;
- mind: from nervous systems to behaviour.

These chapters depend on mathematics that comes later in Part I (real numbers, calculus, linear algebra, probability), so they will be written once that mathematics is in place.

Register

Every notation of the book, grouped by the mathematical model it belongs to, with a link to the definition that introduces it (Rule 1). The register is generated from the definitions themselves.

Logic

Notation	Meaning	Defined in
T, F	the truth values true and false	1.1.1
$\neg P$	not P	1.2.1
$P \wedge Q$	P and Q	1.2.1
$P \vee Q$	P or Q (inclusive)	1.2.1
$P \Rightarrow Q$	if P then Q	1.2.1
$P \Leftrightarrow Q$	P if and only if Q	1.2.1
$A \equiv B$	A and B are logically equivalent	1.2.3
$P(x)$	a statement in which the variable x occurs; not a map value (Rule 7)	1.4.1
$\forall x\ P(x)$	for all x, $P(x)$	1.4.2
$\exists x\ P(x)$	there is an x with $P(x)$	1.4.2
$\exists! x\ P(x)$	there is exactly one x with $P(x)$	1.4.2
$a = b$	a and b are the same object	1.4.3
$a \neq b$	$\neg(a = b)$	1.4.3
$A :\Leftrightarrow B$	the statement A is defined to mean B	1.4.4
$a \coloneqq t$	the name a is defined to be t	1.4.4
$\forall x \in A\ P(x)$	for all x in A, $P(x)$	1.4.7
$\exists x \in A\ P(x)$	there is an x in A with $P(x)$	1.4.7

Sets

Notation	Meaning	Defined in
$x \in A$	x is an element of A	2.1.1
$x \notin A$	$\neg(x \in A)$	2.1.1
$A \subseteq B$	A is a subset of B	2.2.2
$A \subsetneq B$	$A \subseteq B$ and $A \neq B$	2.2.2
$\varnothing$	the empty set	2.3.3
$\{x \in A \mid P(x)\}$	the set of elements of A satisfying P	2.4.1
$\{a, b\}$	the set with exactly the elements a and b	2.5.1

Notation	Meaning	Defined in
$\{a\}$	the set with exactly the element a	2.5.1
$\bigcup \mathcal{F}$	the set of elements of elements of $\mathcal{F}$	2.5.2
$A \cup B$	union of A and B	2.5.3
$A \cap B$	intersection of A and B	2.5.3
$A \setminus B$	elements of A not in B	2.5.3
$\bigcap \mathcal{F}$	elements of every element of $\mathcal{F}$, for $\mathcal{F} \neq \varnothing$	2.5.3
$\{a, b, c\}$, ...	$\{a, b\} \cup \{c\}$, and so on	2.5.3
$\mathcal{P}A$	the set of all subsets of A	2.5.6

Relations and orders

Notation	Meaning	Defined in
(a, b)	ordered pair with first component a and second component b	3.1.1
(a, b, c), ...	tuple $((a, b), c)$, and so on	3.1.3
$A \times B$	the set of pairs (a, b) with $a \in A$, $b \in B$	3.2.2
$\{(x, y) \in A \times B \mid P(x, y)\}$	the set of pairs in $A \times B$ satisfying P	3.2.2
$x\ R\ y$	x is related to y by R, i.e. $(x, y) \in R$	3.3.1
$\operatorname{dom} R$, $\operatorname{ran} R$	domain and range of R	3.3.1
R^{-1}	inverse relation: $y\ R^{-1}\ x \Leftrightarrow x\ R\ y$	3.3.1
id_A	identity on A: $x\ \mathrm{id}_A\ y \Leftrightarrow x = y$	3.3.2
$[a]$	equivalence class of a (the relation is fixed by the context)	3.4.1
$A/{\sim}$	set of equivalence classes of $\sim$	3.4.1
$x \leq y$	a partial order	3.5.1
$x < y$	$x \leq y$ and $x \neq y$	3.5.1
$\sup S$, $\inf S$	least upper bound, greatest lower bound of S	3.5.3

Maps

Notation	Meaning	Defined in
$f \colon A \to B$	f is a map from A to B	4.1.1
$x \overset{f}{\longmapsto} y$	the map f sends x to y; the only notation for values of maps	4.1.1
B^A	the set of maps from A to B	4.1.4
$R[X]$	image of the set X under the relation R; for a map f, $f[X]$ is the image and $f^{-1}[Y]$ the preimage	4.2.1
$f\|_X$	restriction of f to X	4.2.1
$S \circ R$	composition: first R, then S	4.4.1
$x \overset{f}{\longmapsto} y \overset{g}{\longmapsto} z$	$x \overset{f}{\longmapsto} y$ and $y \overset{g}{\longmapsto} z$	4.4.3
$a * b$	value of the binary operation $*$ at (a, b); operations are the only maps whose values are terms	4.6.1

Natural numbers

Notation	Meaning	Defined in
$\mathbb{N}$	the set of natural numbers	5.1.3
$0, 1, \ldots, 9$	the first ten natural numbers; 0 is the empty set regarded as a number (Rule 8)	5.1.3
S	the successor map $\mathbb{N} \to \mathbb{N}$, sending n to $n+1$	5.2.3
$m + n$	sum of m and n	5.4.1
$m \cdot n$	product of m and n; $\cdot$ is evaluated before $+$	5.5.1
$m < n$, $m \leq n$	order of natural numbers: $m \in n$, and $m \in n \vee m = n$	5.6.1

Commutative rings and fields

Notation	Meaning	Defined in
$(R, +, \cdot, -, 0, 1)$	a commutative ring	6.2.3
x^{-1}	multiplicative inverse of $x \neq 0$ in a field	7.2.3
x/y	$x \cdot y^{-1}$	7.2.3

Integers

Notation	Meaning	Defined in
$\mathbb{Z}$	the set of integers	6.1.3
$[(a, b)]$	the integer "$a - b$"; replaced by $a - b$ after Theorem 6.3.1	6.1.3
$x + y$, $x \cdot y$	sum and product of integers (extend those of $\mathbb{N}$)	6.2.1
$-x$	the negative of x	6.2.1
0, 1	$[(0, 0)]$ and $[(1, 0)]$ (identified with $0, 1 \in \mathbb{N}$)	6.2.1
$x \leq y$, $x < y$	order of integers: $y - x \in \mathbb{N}$ (extends the order of $\mathbb{N}$)	6.5.1

Rational numbers

Notation	Meaning	Defined in
$\mathbb{Z}^*$	the nonzero integers	7.1.1
$\mathbb{Q}$	the set of rational numbers	7.1.3
$[(a, b)]$	the rational number "a over b"; replaced by a/b after Theorem 7.3.1	7.1.3
$x + y$, $-x$, $x \cdot y$, 0, 1	the ring operations of $\mathbb{Q}$ (extend those of $\mathbb{Z}$)	7.2.1
$x \leq y$, $x < y$	order of rationals: $y - x = a/b$ with $a, b \in \mathbb{N}$ (extends the order of $\mathbb{Z}$)	7.4.1